

Made, Earned, or Extracted? Platform Scale, Housing Scarcity, and Short-Term Rental Regulation

Diego A. Diaz

Universidad del Desarrollo

June 2026

Abstract

Digital platforms create surplus by matching users, but in scarce-asset markets they can also monetize externally produced scarcity. I study short-term rentals using NYC Airbnb listings, Zillow metro panels, NYC Local Law 18 registrations, council-district boundaries, and ACS housing denominators. High-availability whole-home listings in 2019 were concentrated among multi-listing hosts and high-rent districts, with posted capacity prices far above local rent benchmarks. After New York’s platform-verification regime, the legal registry was much smaller and spatially distinct. Metro policy-window estimates and synthetic controls suggest rent relief after some restrictions, but heterogeneous timing and pretrends bound the causal claim.

JEL Codes: R31, R52, L86, D62, L51.

Keywords: short-term rentals, Airbnb, housing scarcity, platform markets, regulation, rent extraction, urban economics.

1 Introduction

A central question in modern political economy is how to distinguish wealth created by innovation from wealth obtained by controlling scarce assets, regulatory bottlenecks, or market positions. Digital platforms sharpen the question because they often combine both forces. A platform can lower search costs, increase utilization of idle capacity, and generate genuine consumer and producer surplus. It can also amplify scarcity by connecting a scarce local asset to a much larger willingness-to-pay pool. When that asset is housing, the distinction is not merely semantic. A spare bedroom rented during a holiday weekend is a different

economic object from an apartment removed from the long-term rental market and sold repeatedly to tourists. The platform transaction looks similar in each case, but the welfare content differs.

This paper studies that distinction in the short-term rental market. The empirical setting is housing, and the motivating platform is Airbnb, but the broader economic question is about the composition of platform surplus. When a platform scales in a market with fixed or inelastic local supply, how much of its private value comes from better matching and how much comes from redirecting a scarce asset toward users with higher willingness to pay? The answer matters for welfare analysis, antitrust, local regulation, and the measurement of entrepreneurial returns. A platform that unlocks idle capacity deserves a different regulatory treatment from a platform that reallocates permanent housing toward transient demand while leaving the external cost on incumbent renters.

I use the short-term rental market for three reasons. First, the relevant asset is locally scarce. Housing supply is not perfectly elastic, especially in high-income coastal metropolitan areas where geography, land-use regulation, and political constraints limit new construction [Saiz, 2010, Gyourko et al., 2008, Hsieh and Moretti, 2019]. Second, the platform’s activity can be divided into economically distinct margins. A private room in an occupied home is plausibly a use of excess capacity. A whole apartment available most of the year is more plausibly a hotel-like use of residential space. Third, local governments have adopted registration rules, primary-residence restrictions, platform-verification requirements, and enforcement systems, generating variation in the legal ability to supply short-term rentals. These features make short-term rentals a useful case for studying the boundary between innovation rents and scarcity rents.

The paper makes two contributions. The first is conceptual. I propose a surplus decomposition for platform activity in scarce-asset markets. Let platform-generated private value be written as the sum of matching surplus, scarcity rents, regulatory rents, and network rents, net of external costs. Matching surplus arises when the platform allows otherwise underused housing to be rented to travelers at low transaction cost. Scarcity rents arise when the platform diverts housing from long-term residential use in a market where the marginal resident faces an inelastic supply curve. Regulatory rents arise when platform activity exploits differences between residential and hotel regulation, tax rules, safety rules, or tenant protections. Network rents arise from liquidity, reputation, data, and platform scale. This decomposition converts the normative distinction between wealth that is “made” and wealth that is “earned” into empirically observable margins: type of unit, availability, host portfolio size, regulatory status, and housing-market response.

The second contribution is empirical. I combine listing microdata, housing-market time

series, and post-enforcement regulatory data. The listing analysis uses 48,884 New York City Airbnb listings from the public AB_NYC_2019 file, with 37,455 unique host identifiers. The housing-market analysis uses Zillow Observed Rent Index and Zillow Home Value Index data for 146 large U.S. metropolitan areas from 2015-01-31 through 2026-03-31. The regulatory analysis uses New York City’s Short-Term Rental Registration Report for the fiscal year ending June 30, 2025, which documents the implementation of Local Law 18. The spatial accounting uses ACS tract housing and gross-rent estimates assigned to current council districts. All reported estimates and descriptive statistics come from these public data sources.

The empirical results are organized around three objects. The first is a listing-level exposure measure. I classify a listing as a potential housing-reallocation candidate if it is an entire home or apartment, is available at least 180 days per year, and has a minimum stay of 30 nights or fewer. The threshold is intentionally conservative: it does not prove that the unit would otherwise have housed a long-term resident, but it identifies listings whose observable characteristics are inconsistent with purely incidental home sharing. In the 2019 New York City sample, 52.0 percent of listings are entire homes or apartments, 33.9 percent are operated by hosts with multiple listings, and 7,509 listings meet the high-availability whole-home screen. The high-availability whole-home segment accounts for 15.4 percent of all listings and a much larger share of the gross capacity-value measure because nightly prices are higher in whole-unit listings.

The second object is market organization. Host concentration is not extreme in the sense of a few hosts controlling the market: the listing-count Gini coefficient across hosts is 0.22, and the top 100 host identifiers account for 6.2 percent of listings. Yet the distribution matters because multi-listing hosts operate 48.0 percent of the high-availability whole-home segment, and hosts with at least five listings operate 27.7 percent of that segment. This pattern supports a distinction between casual hosts and professional or semi-professional suppliers. The economic incidence of platform activity depends on that distinction. A city may rationally allow spare-room sharing while restricting non-owner-occupied whole-unit rentals, because the two activities have different effects on residential supply.

The third object is the housing-market response around restrictions. I estimate a stacked event-study design around three restrictive policy events: Boston’s short-term-rental ordinance in January 2019, Los Angeles’s Home-Sharing Ordinance in July 2019, and New York City’s Local Law 18 enforcement in September 2023. The design compares treated metros with large untreated metros, uses stack-specific unit and month fixed effects, and reports dynamic estimates relative to the month before treatment. I also estimate city-specific synthetic controls that match each treated metro’s pre-policy Zillow trajectory with weighted untreated

donor metros [Abadie et al., 2010, 2015]. The identifying variation is deliberately coarse. Policy adoption is not random, treatment is heterogeneous, and city-level restrictions can coincide with other housing-market shocks. The event study and synthetic-control checks therefore discipline the timing and heterogeneity of aggregate market outcomes rather than replacing tract-border or listing-level enforcement evidence. The stacked full-sample estimates show rents in treated metros -3.4 percent lower 13–24 months after the event, with a cluster-robust standard error of 1.7 percentage points, but the same coefficient becomes 1.6 percent after excluding March 2020 through December 2021 and is then based on only 13 treated event-months. Home values are 0.9 percent higher over the same horizon in the full sample, with a standard error of 2.0 percentage points. The rent evidence is therefore suggestive and heterogeneous rather than a stable pooled causal estimate; the home-value evidence is too imprecise and directionally mixed to support a simple asset-price story.

New York’s Local Law 18 provides an institutional check on the listing evidence. The law requires short-term rental hosts to register with the Office of Special Enforcement and requires platforms to verify registration before processing transactions. The FY2025 report records 2,952 active registrations as of June 30, 2025, 1,582 applications submitted during the fiscal year, 640 granted applications, and 471 denied applications. It also reports 4,329 total denied applications through June 30, 2025, including 553 denials involving rent-regulated units. The registry is much smaller than the pre-enforcement listing market observed in the 2019 data, and it is targeted at a different legal object: host-present, compliant short-term rental activity rather than the broad set of platform listings that included whole-unit rentals. The contrast illustrates how formal regulation can separate legal home sharing from the housing-reallocation segment.

The analysis relates to a growing empirical literature on short-term rentals and housing markets. Barron et al. [2021] estimate that Airbnb activity raises rents and house prices, with mechanisms consistent with housing reallocation. Koster et al. [2021] use regulatory variation in Los Angeles and find that restricting short-term rentals affects housing prices. Garcia-López et al. [2020] document housing-market effects in Barcelona. These papers establish that short-term rental platforms can affect local housing markets. My contribution is not to replicate a single causal estimate with superior microgeography. Instead, I connect that literature to a decomposition of platform surplus and to the regulatory distinction between casual and professionalized supply. The welfare question is not simply whether short-term rentals raise rents. It is which part of the platform market produces value through matching and which part produces returns by bidding residential space away from long-term residents.

The paper also relates to work on peer-to-peer markets and platform welfare. Einav

et al. [2016] emphasize that peer-to-peer platforms reduce transaction costs and enable decentralized supply. Farronato and Fradkin [2022] estimate welfare gains from Airbnb entry in accommodation markets, especially where hotel supply is constrained. That welfare logic is important. A short-term rental platform can reduce hotel market power, expand travel options, and generate income for households. But the same hotel-supply constraint that makes entry valuable to travelers can make residential spillovers costly when the entrant’s supply is drawn from the housing stock. The welfare problem is therefore one of incidence and margin. The platform can be welfare-enhancing overall while still generating negative externalities in the specific segment of high-availability whole-unit listings.

The remainder of the paper proceeds in six steps. I first locate the contribution relative to the literatures on short-term rentals, platform markets, housing supply, and local regulation. I then develop the surplus decomposition, describe the data and institutional setting, define the empirical objects, report the listing and registration evidence, estimate housing-market dynamics around restrictions, and interpret the results for welfare and policy.

2 Related Literature and Contribution

This paper sits at the intersection of four literatures that usually speak to each other only partially. The first is the empirical literature on short-term rentals and housing markets. The canonical question in that literature is whether Airbnb activity raises rents or house prices. The evidence generally points to positive effects in markets where listings are numerous and housing supply is constrained, although magnitudes depend on geography, measurement, and identification. Barron et al. [2021] use U.S. data and an identification strategy based on Airbnb penetration interacted with tourism demand to show that growth in Airbnb listings is associated with higher rents and home prices. Their mechanism is especially relevant here: they emphasize that short-term rental activity can reduce the long-term rental supply available to residents. Koster et al. [2021] use quasi-experimental variation from Los Angeles regulations and show that restrictions on home sharing can affect nearby housing prices. Garcia-López et al. [2020] study Barcelona and document effects on rents, transaction prices, and neighborhood composition. Taken together, these papers establish that short-term rentals can matter for local housing markets. They do not by themselves answer how the private surplus of the platform should be decomposed across innovation, scarcity, and regulatory margins.

The second literature studies peer-to-peer platforms as institutions that lower transaction costs. In the standard platform view, entry by peers can expand supply, discipline incumbents, and create welfare gains for consumers. Einav et al. [2016] describe peer-to-peer markets

as mechanisms for decentralized exchange in which digital reputation, search, and payment systems permit small suppliers to transact with dispersed consumers. Farronato and Fradkin [2022] estimate the welfare effects of Airbnb entry and show that peer supply can generate substantial gains in accommodation markets, particularly where hotel capacity is constrained. That result is central rather than incidental to this paper. If Airbnb generated no matching surplus, the policy problem would be straightforward. The difficult economic question is precisely that the platform can create traveler surplus and host surplus while also using residential housing as an input. Hotel-supply constraints make peer entry valuable for travelers; housing-supply constraints make the source of peer supply consequential for residents.

The third literature is urban economics on housing supply. Housing is an asset with local market-clearing, durable capital, regulatory constraints, and spatial heterogeneity. Saiz [2010] shows that geography affects housing supply elasticity, and Gyourko et al. [2008] document variation in local regulatory constraints. Hsieh and Moretti [2019] link housing restrictions to spatial misallocation. These papers are not about short-term rentals, but they are necessary for interpreting short-term rental effects. In an elastic housing market, the conversion of some units to visitor accommodation induces new construction or substitution with limited rent pressure. In an inelastic market, the same conversion can have a larger incidence on renters. The relevant short-term rental question is therefore conditional: the welfare effect depends on the interaction between platform demand and the elasticity of local housing supply. A national average effect can conceal precisely the markets in which the welfare stakes are highest.

The fourth literature concerns regulation, market power, and rent extraction. Two-sided-market models emphasize that platform scale can create network value and platform market power [Rochet and Tirole, 2003, Rysman, 2009]. Urban regulation creates an additional source of rents. Hotels face taxes, safety rules, zoning rules, labor regulation, and often political scrutiny. Long-term residential landlords face tenant protections, rent regulation in some jurisdictions, habitability rules, and eviction constraints. Short-term rental platforms emerged by connecting residential units to visitor demand under legal categories that were often designed before the platform existed. That institutional mismatch is a classic source of regulatory arbitrage. A unit may be residential for purposes of zoning and tenant protection but commercial for purposes of actual use. The resulting private return is not purely technological. It also reflects the ability to arbitrage rules across regulatory domains.

The contribution of this paper is to connect these literatures through an accounting framework and then discipline that framework with public data. The short-term rental literature already asks whether listings affect rents. The platform literature already asks whether peer entry creates welfare. The urban literature already emphasizes housing supply

constraints. The regulation literature already emphasizes arbitrage and enforcement. What is missing is a common object that links them: the decomposition of platform value in a scarce-asset market. The paper’s object is not simply the sign of a rent coefficient. It is the mapping from listing types, host organization, regulation, and housing-market dynamics to components of surplus.

This framing also changes the interpretation of heterogeneity. In much empirical work, heterogeneity is a robustness exercise: effects are larger here, smaller there. In this paper, heterogeneity is part of the theory. A private room in an owner-occupied dwelling is not a noisy version of a whole-unit conversion. It is a different economic margin. A low-availability listing does not merely have a smaller treatment dose; it represents a different opportunity cost. A multi-listing host is not just a host with a larger portfolio; the portfolio signals that the activity may be organized as business supply rather than incidental household supply. A neighborhood with high tourism demand and low rental vacancy is not just a high-exposure submarket; it is the place where the shadow price of residential space is most likely to be capitalized by the platform.

This paper therefore makes a relatively conservative empirical claim in service of a broader conceptual claim. I do not argue that short-term rentals caused the housing crisis. The dominant sources of high housing costs in constrained metropolitan areas are long-run supply restrictions, land scarcity, demand growth, amenities, incomes, and macro-financial conditions. Nor do I argue that platform entry is socially wasteful. The evidence in the platform literature makes clear that peer entry can create welfare. The paper’s claim is narrower and sharper: once a short-term rental platform scales beyond occasional home sharing, the source of platform surplus depends on whether the marginal transaction uses idle housing capacity or reallocates scarce residential units. That distinction is observable enough to guide empirical work and policy.

The paper is also distinct from a purely legal or policy analysis of short-term rental ordinances. Local governments often justify regulation in terms of nuisance, hotel competition, tax compliance, or neighborhood character. Those considerations are real, but the central economic issue is the opportunity cost of the dwelling unit. If the host is present and the unit remains a residence, the opportunity cost is spare capacity and inconvenience. If the host is absent and the unit is available most of the year, the opportunity cost is the long-term residential use of a scarce asset. The registration and verification systems in New York, Boston, and Los Angeles can be interpreted as institutional attempts to make that opportunity cost visible and enforceable.

3 Conceptual Framework

The starting point is a local housing market with two uses for a dwelling unit: long-term residential occupancy and short-term visitor accommodation. A platform reduces the cost of matching a host with a visitor. In a frictionless model with idle rooms and no spillovers, the welfare effect is straightforward. The platform creates a new transaction between a guest whose willingness to pay exceeds the host's opportunity cost and a host who would otherwise leave the room unused. Consumer surplus rises, host income rises, and local housing supply for residents is unchanged. This is the pure matching view of short-term rentals.

The pure matching view becomes incomplete when the platform can also serve whole residential units. Let H denote the local stock of housing units, L long-term residential use, and S short-term rental use, with $H = L + S$ in the short run. Let residents face inverse demand $P_L(L)$ and visitors face inverse demand $P_S(S)$ for local short-term accommodation. A platform increases the effective price received for short-term use by lowering search costs, reducing payment risk, and increasing the set of visitors. If the short-term rental return net of costs exceeds the long-term rental return, some units move from L to S . The private host gain is the difference between short-term and long-term returns. The external cost is the effect of lower L on resident rents and displacement risk.

The platform's private surplus can be written as

$$\Pi^P = M + R^H + R^G + R^N, \quad (1)$$

where M is matching surplus from idle or underutilized capacity, R^H is a housing-scarcity rent from reallocating scarce residential space, R^G is a regulatory rent from operating under a different regulatory burden than hotels or long-term landlords, and R^N is a network rent from platform liquidity, data, brand, and reputation. Local welfare is

$$W = CS_T + PS_H + \tau - EC_R - EC_N - C_E, \quad (2)$$

where CS_T is traveler consumer surplus, PS_H is host surplus, τ is tax revenue, EC_R is the external cost borne by renters through higher rents or lower availability, EC_N captures neighborhood costs such as noise and congestion, and C_E is the administrative and compliance cost of enforcement.

The decomposition is useful because the components map to observable margins. Matching surplus is associated with private rooms, shared rooms, low-availability listings, or listings in owner-occupied units. Housing-scarcity rents are associated with entire homes, high availability, low minimum-night requirements, and concentration in high-rent or high-tourism

neighborhoods. Regulatory rents are associated with listings that would be unlawful under hotel, zoning, rent-regulation, occupancy, or registration rules. Network rents are harder to observe directly, but host concentration and platform verification rules provide indirect evidence about scale and market structure.

The key empirical challenge is that observed platform activity is an equilibrium object. Listings are more common in neighborhoods with high tourist demand, high amenities, and high rents. Those neighborhoods may also experience rent growth for reasons unrelated to short-term rentals. A naive regression of rents on listings would therefore combine platform effects with omitted local demand. The literature addresses this challenge through instrumental variables, border designs, and policy variation [Barron et al., 2021, Koster et al., 2021, Garcia-López et al., 2020]. This paper uses a complementary strategy. It does not claim that a cross-section of listings identifies causal rent effects. Instead, it uses listing attributes to classify the economic margin of platform supply and uses policy events to examine aggregate housing-market dynamics around restrictions.

The decomposition also clarifies why policy can be targeted rather than binary. A city does not need to decide that all platform activity is innovation or all platform activity is extraction. It can regulate by margin. For example, it can allow home sharing in primary residences, require registration and tax remittance, prohibit short-term rentals in rent-regulated or subsidized units, restrict entire-home rentals, cap rental nights, or require platforms to verify registration. These policies are not arbitrary; they correspond to components of the surplus decomposition. A primary-residence rule attempts to preserve M while reducing R^H . A registration and verification rule attempts to reduce R^G . A night cap attempts to distinguish occasional use from conversion. A tax rule attempts to recover some of the external cost through public revenue.

4 Data and Institutional Setting

The empirical analysis uses public source files covering listings, housing-market outcomes, regulatory registrations, political geography, and housing-stock denominators. The first source is the AB_NYC_2019 listing file, a cross-section of New York City Airbnb listings in 2019 [New York City Airbnb Open Data, 2019]. The file contains 48,884 observations after removing zero-price listings and 37,455 unique host identifiers. It reports listing identifiers, host identifiers, borough, neighborhood, latitude, longitude, room type, nightly price, minimum nights, number of reviews, reviews per month, the host’s reported number of listings, and days available during the year. I winsorize nightly price at the 99th percentile for summary statistics and price regressions to prevent extreme listing prices from dominating

capacity-value measures. The main exposure classification uses only room type, availability, and minimum nights.

The second source is Zillow Research housing-market time-series data [Zillow Research, 2026]. I use the Zillow Observed Rent Index (ZORI) and the Zillow Home Value Index (ZHVI) at the metropolitan level. ZORI is a repeat-rent, dollar-denominated measure of typical observed market rent; ZHVI measures typical home values. I reshape the public CSV files into a metro-month panel, retain metropolitan statistical areas with at least 90 non-missing ZORI months, and focus on the 150 largest metros by Zillow size rank. The resulting panel contains 19,710 metro-month observations for 146 metros between 2015-01-31 and 2026-03-31. The event-study sample stacks policy windows around three treated metros and excludes other treated metros from the control group within each stack.

The third source is New York City’s FY2025 Short-Term Rental Registration Report [New York City Mayor’s Office of Special Enforcement, 2025]. Local Law 18 established registration requirements for short-term residential rentals and verification requirements for booking platforms. The report covers July 1, 2024 through June 30, 2025 and provides council-district counts of active registrations, application statuses, processing times, denial reasons, and enforcement information. I parse the workbook directly in the replication code. The report records 2,952 active registrations as of June 30, 2025 and 3,111 registrations active at any time during the fiscal year. It also records 1,582 applications submitted during the fiscal year, of which 640 were granted and 471 were denied as of June 30, 2025.

The fourth source is the NYC Open Data City Council Districts GeoJSON file [NYC Open Data and New York City Department of City Planning, 2026]. I use the official polygons to assign listing coordinates to current council districts. This creates a common spatial unit for comparing the pre-enforcement listing market with the post-enforcement registration report. The assignment covers 99.97 percent of 2019 listings.

The fifth source is the 2019–2023 American Community Survey five-year detailed-table Summary File [U.S. Census Bureau, 2024a]. I extract tract-level housing units, occupied and vacant units, tenure, and median gross rent for the five New York City counties. I then use ACS 2023 TIGERweb tract internal points [U.S. Census Bureau, 2024b] to assign 2,327 census tracts to council districts. The point-in-polygon assignment covers 2,324 tracts; a nearest-boundary fallback within one kilometer assigns seven waterfront tracts whose internal points fall just outside council polygons, and the remaining unassigned tracts have zero ACS housing units. This tract-to-district crosswalk provides 3,649,896 housing units, 2,226,896 renter-occupied households, and 336,580 vacant housing units for normalizing short-term-rental exposure.

The institutional setting is shaped by local regulation. Boston’s ordinance took effect

on January 1, 2019 and limited short-term rentals to eligible owner-occupied residential units [City of Boston, 2018]. Los Angeles’s Home-Sharing Ordinance took effect on July 1, 2019 and restricted home sharing to primary residences, with registration requirements and prohibitions for rent-stabilized units [Los Angeles Housing Department, 2019]. New York City’s Local Law 18 began platform-verification enforcement in September 2023 [New York City Mayor’s Office of Special Enforcement, 2023]. These policies share a common economic logic: they try to preserve occasional home sharing while reducing the conversion of residential units into de facto hotel supply.

The data have limitations. The 2019 New York listing file is a cross-section, not a panel of individual listings before and after a policy change. The Zillow data are at the metro level, so they are too aggregated to isolate neighborhood-level effects or border discontinuities. The OSE report measures legal registrations after enforcement, not all attempted or illegal short-term rental activity. The council-district comparison assigns 2019 listing coordinates and ACS tract internal points to current boundaries because those boundaries match the FY2025 reporting geography. These limitations shape the interpretation. The listing analysis identifies the segment where housing reallocation is plausible; it does not estimate how many units were actually removed from long-term rental use. The event study describes housing-market dynamics around restrictions; it does not prove that the restrictions alone caused the observed dynamics. The value of combining the sources is that each disciplines a different piece of the welfare decomposition.

The policy events used in the event study are not interchangeable, but they share a common treatment logic. Boston’s ordinance is coded in January 2019 because that is when the local short-term-rental framework took effect. The ordinance limited eligible units and tied legality to owner occupancy and registration. This design attempts to distinguish residential use from commercial use by asking whether the supplier is also a resident. Los Angeles is coded in July 2019, when its Home-Sharing Ordinance took effect. The Los Angeles policy restricted short-term rental activity to a primary residence, required registration, and excluded units subject to important housing protections. New York is coded in September 2023, when platform-verification enforcement under Local Law 18 began. New York’s approach is especially stringent because it makes platforms responsible for verifying registration before facilitating short-term transactions. In economic terms, these policies are not simple taxes or quantity caps. They change which margins of short-term rental supply can legally operate.

The three policies also differ in how directly they target housing reallocation. A primary-residence requirement separates residents monetizing temporary excess capacity from investors using residential units as visitor accommodation. A registration requirement reduces anonymity and increases the probability that legal status can be verified. A platform-

verification requirement changes enforcement from a host-level after-the-fact process to a transaction-level gatekeeping process. The last feature is important. If enforcement is left entirely to local inspectors after transactions occur, illegal listings may remain profitable because detection is costly and delayed. If the platform must verify eligibility before a booking, the expected return to noncompliant supply falls immediately. That distinction is why the New York registry is informative even though it is not a complete census of attempted short-term rental activity.

The listing, Zillow, OSE, boundary, and ACS data measure different levels of the same economic process. The listing file measures potential supply before strict New York verification. The Zillow data measure aggregate housing-market outcomes around restrictions in several large metros. The OSE workbook measures the regulated market after New York’s verification regime is in place. The council-district crosswalk asks whether the spatial distribution of the old high-exposure segment resembles the spatial distribution of the new legal market, both in raw counts and relative to local renter households. No single source provides the complete causal chain from individual listing to renter welfare. The empirical strategy therefore triangulates. The listing data show the size and composition of the plausibly reallocation-relevant segment. The policy event study asks whether restrictions are followed by changes in rent and value indexes. The registry data show how many legal short-term rental registrations remain after enforcement. The combination is stronger than any one component but still requires careful interpretation.

The data construction follows three principles intended to make the evidence auditable. First, classifications are based on observable rules rather than discretionary coding. The high-availability whole-home measure uses room type, annual availability, and minimum nights. Second, the housing-market estimates use publicly available time series and event dates that are fixed before estimation. Third, all manuscript numbers are written from the replication output through LaTeX macros rather than manually typed into the text. This reduces the risk that revisions to the code leave the manuscript inconsistent with the tables.

5 Empirical Objects and Estimating Equations

The first empirical object is potential exposure to residential reallocation. For listing i , define

$$R_i = \mathbf{1}\{\text{entire home}_i = 1, \text{availability}_i \geq 180, \text{minimum nights}_i \leq 30\}. \quad (3)$$

This indicator is not a causal estimand. It is an accounting screen. A listing with $R_i = 1$ is not necessarily a converted long-term rental unit, but it has the observable features most

consistent with a unit that could substitute for long-term occupancy. Conversely, a private room or an entire home available only a few days a year is more likely to reflect occasional home sharing. I compute exposure shares by borough and neighborhood and construct a gross annual capacity-value proxy,

$$CV_i = p_i^w \times a_i, \quad (4)$$

where p_i^w is winsorized nightly price and a_i is reported annual availability. This is not realized revenue because the data do not include occupancy or transaction prices. It is a capacity-value measure: the gross value the listing could generate if available nights were booked at the listed price. The measure is useful for comparing the economic scale of listing types, not for estimating host income.

The second object is host organization. I compute the share of listings operated by hosts with more than one listing, the share operated by hosts with at least five listings, the listing share of the top 10 and top 100 host identifiers, and the Gini coefficient of listing counts across hosts. The goal is to separate atomistic home sharing from portfolio supply. Concentration matters for welfare because multi-listing hosts are less likely to represent households monetizing occasional spare capacity and more likely to represent a business model built around repeated use of residential units.

The third object is a listing-level price equation. I estimate

$$\begin{aligned} \log p_i = & \alpha + \beta_1 \text{PrivateRoom}_i + \beta_2 \text{SharedRoom}_i + \beta_3 \text{MultiHost}_i \\ & + \beta_4 \text{ProfessionalHost}_i + X_i' \gamma + \delta_b + \varepsilon_i, \end{aligned} \quad (5)$$

where X_i includes log minimum nights, log reviews, reviews per month, and availability; δ_b are borough fixed effects. This regression is descriptive. It shows how prices covary with listing type and host organization after conditioning on broad geography, but it does not identify causal price effects.

The fourth object is the stacked metro event study. For each treated metro g and policy month E_g , I create a stack containing the treated metro and untreated control metros observed within a 24-month window around E_g . Let $k = t - E_g$ denote event time. I estimate

$$Y_{gst} = \mu_{gs} + \lambda_{st} + \sum_{k=-24}^{24} \beta_k [\mathbf{1}\{g \text{ treated in stack } s\} \times \mathbf{1}\{t - E_g = k\}] + u_{gst}, \quad (6)$$

where Y_{gst} is log ZORI or log ZHVI, μ_{gs} are stack-by-metro fixed effects, and λ_{st} are stack-by-month fixed effects. Month $k = -1$ is omitted. Standard errors are clustered by metro. I also report binned coefficients for months -24 to -13, -12 to -2, month 0, months 1 to 12, and

months 13 to 24.

Because the pre-period diagnostics are important, I also compute a linear-pretrend-adjusted diagnostic from the monthly event-study coefficients. Let $\hat{\beta}_k$ denote the monthly coefficient for event month k . I estimate a pre-period differential trend by fitting

$$\hat{\beta}_k = \delta(k + 1) + e_k, \quad k = -24, \dots, -2, \quad (7)$$

so that the omitted month $k = -1$ is the anchor. For a post-period bin B , I report

$$\hat{\theta}_B^{adj} = \frac{1}{|B|} \sum_{k \in B} \hat{\beta}_k - \hat{\delta} \frac{1}{|B|} \sum_{k \in B} (k + 1). \quad (8)$$

This adjustment asks whether the post-period gap remains after subtracting the linear differential trend implied by the monthly pre-treatment coefficients. It is a sensitivity diagnostic, not a substitute for the identifying assumption that policy timing is as-good-as-random conditional on the fixed effects.

As a complementary design, I estimate synthetic controls for each treated metro and outcome. I normalize each treated and donor metro to the month before policy implementation, screen the untreated donor pool to the 20 closest pre-period trajectories, and estimate nonnegative donor weights that sum to one using months -24 to -2. The synthetic-control gap is the treated metro’s normalized log outcome minus the weighted donor outcome. This check asks whether the pooled event-study pattern survives when each treated market is compared with its own pre-period-matched donor combination rather than with a common stacked control mean.

This event-study design is deliberately modest. Modern difference-in-differences research shows that treatment-effect heterogeneity and staggered timing can invalidate simple two-way fixed effects estimators [Goodman-Bacon, 2021, Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, de Chaisemartin and D’Haultfoeuille, 2020, Borusyak et al., 2024]. Stacking events reduces contamination from already-treated controls, but it does not solve endogenous policy adoption or simultaneous local shocks. Synthetic controls improve pre-period fit for each treated metro, but they still require the weighted donor path to approximate the treated counterfactual after the policy date. For that reason, I interpret the estimates as policy-window evidence. A stronger design would use tract-level or parcel-level data, border discontinuities, detailed enforcement dates, and listing-level measures of regulatory compliance.

6 Identification Logic

The identification strategy is best understood as a sequence of increasingly demanding empirical objects rather than as a single claim that all estimates are causal in the same sense. The first set of results is an accounting exercise. It asks what share of the observed listing market lies on a margin where housing reallocation is plausible. The identifying assumption is minimal because no causal effect is attached to the classification. The key risk is measurement error: some high-availability whole-home listings are not converted long-term rentals, and some lower-availability listings may still affect residential supply. I address that risk by describing the screen as an exposure measure and by reporting the underlying listing attributes rather than a binary legal classification.

The second set of results is descriptive market organization. Host concentration and multi-listing shares are informative because they identify whether supply is dispersed across households or organized in portfolios. The identifying assumption is again modest: the host identifier in the listing file must consistently identify hosts within the sample. It need not identify legal ownership, beneficial ownership, or property-management relationships. The empirical object is not ownership concentration in the legal sense. It is platform-supply concentration as observed by users and by the platform. That distinction matters because a property manager operating many listings under one identifier can shape platform supply even if the underlying owners differ.

The third set of results is a descriptive price model. The log-price regression conditions on room type, host organization, review variables, availability, minimum nights, and borough fixed effects. The coefficients show how the market prices listing attributes in the cross-section. They should not be interpreted as hedonic estimates of causal willingness to pay because unobserved location quality, property quality, and host strategy remain important. Their role in the paper is narrower: they show that the high-availability whole-unit segment is economically valuable, so the reallocation margin is not a small or low-price residual.

The fourth set of results is the stacked event study around policy restrictions. The causal interpretation would require that, after stack-specific unit and time effects, treated metros would have followed the same rent and home-value path as control metros absent the restriction. That assumption is demanding. Boston, Los Angeles, and New York regulate short-term rentals partly because they are high-demand cities with active housing politics. The event-study evidence therefore cannot by itself settle the causal effect of regulation. Its value is that it examines whether the timing of restrictions is consistent with the mechanism and whether pre-period dynamics are obviously inconsistent with a parallel-trends interpretation.

Several design choices are intended to reduce, but not eliminate, threats to identification.

I stack policy events so that each treated metro is compared with untreated large metros in its own event window. Other treated metros are excluded from the control group within a stack to avoid comparing a treated market with a market that is itself undergoing a similar restriction. The specification includes stack-by-metro fixed effects, which absorb time-invariant level differences, and stack-by-month fixed effects, which absorb shocks common to all metros within a policy window. Standard errors are clustered by metro. The binned table reports pre-period coefficients because pretrends are more informative for credibility than a single post-period estimate. I also report formal joint tests of the monthly pre-treatment coefficients, a linear-pretrend-adjusted post-period diagnostic using the same residualized event-study design and metro-clustered covariance matrix, a matched-control sensitivity that restricts each stack to untreated metros with the closest pre-policy Zillow trajectories, an event-stack contribution diagnostic that decomposes the residualized pooled coefficient by treated policy episode, and a placebo-triplet diagnostic that reassigns the three real policy dates to untreated metros before re-estimating the stacked binned specification. The synthetic-control exercise provides an additional diagnostic: it shows whether the pooled estimate masks city-specific trajectories that move in different directions after matching pre-period outcomes.

The design remains conservative relative to an ideal top-field empirical design. A stronger design would use within-city variation in legal eligibility, such as building-level rent-regulation status, owner-occupancy status, or borders between jurisdictions with different ordinances. It would observe listings before and after enforcement, link listings to parcels or housing units, and measure long-term rental supply directly. It would also exploit the date at which platform verification prevents bookings rather than merely the month in which a local law is coded at the metro level. The public data assembled here do not contain all of those features. Rather than hide that limitation, the paper makes the strength of each empirical object explicit.

This explicitness is important for interpreting the estimates. If the event study finds that rents are lower after restrictions, the result is consistent with housing-supply relaxation but could also reflect differential local shocks. If the event study finds no effect, that could mean short-term rentals are too small to matter at the metro level, that enforcement was weak, that effects are concentrated in neighborhoods not visible in metro averages, or that the relevant housing market adjusted through prices rather than rents. If the event study finds positive rent effects after restrictions, that could reflect endogenous regulation in cities facing worsening housing pressure. The paper’s contribution is therefore not a single decisive coefficient. It is the integration of listing composition, policy enforcement, and housing-market dynamics into one welfare framework.

The framework also clarifies why standard difference-in-differences concerns are substan-

tively meaningful here. Treatment-effect heterogeneity is not a nuisance parameter; it is a prediction. Restrictions should matter more where short-term rental supply is professionalized, where whole-unit listings are common, where tourist demand is high, and where housing supply is inelastic. A pooled event-study coefficient averages across treated cities whose regulations, market conditions, and enforcement intensities differ. For that reason, the estimates are presented alongside treated-market summaries and New York registry evidence rather than as a stand-alone national effect.

7 New York City Listing Evidence

Table 1 reports the composition of the 2019 New York City listing market. Entire homes or apartments account for 52.0 percent of listings, private rooms account for 45.7 percent, and shared rooms make up the small residual. The median nightly price is \$106. The key distinction is between listing type and availability. Entire homes are not automatically conversions from long-term housing. A household may rent an apartment while away for a short period. But an entire-home listing with high annual availability is economically closer to hotel supply than to incidental home sharing.

Table 1: NYC Airbnb listing composition by listing type, 2019

Listing type	Listings	Share (%)	Median price	Median days available	High-availability share (%)	Multi-listing share (%)
Entire home/apt	25407	51.974	160.000	42.000	30.598	25.241
Private room	22319	45.657	70.000	45.000	28.182	42.354
Shared room	1158	2.369	45.000	90.000	42.228	61.917

Notes: Listing-level sample is the public AB_NYC_2019 file. Prices are winsorized at the 99th percentile for summary statistics. High-availability equals at least 180 available days in the calendar-year availability field.

The exposure screen identifies 7,509 high-availability whole-home listings. This is a large enough segment to matter in neighborhoods with tight rental markets, even though it is not the majority of all listings. The main definition also accounts for 56.9 percent of listing capacity value, reflecting both high availability and higher whole-unit prices. Because any cutoff can appear arbitrary, Table 2 varies the availability threshold and minimum-stay cap. A looser 90-day availability threshold with the same 30-night cap identifies 10,172 listings, while a stricter 270-day threshold identifies 4,755 listings. The exact count changes, but the qualitative fact does not: thousands of entire-home listings were available enough to resemble a housing-reallocation margin rather than incidental home sharing.

Table 3 adds a review-based activity diagnostic. The exposure screen is based on unit type, annual availability, and minimum-stay rules; it does not by itself prove that a listing was frequently booked or that the unit otherwise would have housed a long-term renter. The

Table 2: Sensitivity of potential housing-reallocation exposure definitions

Definition	Candidate listings	Share of all listings (%)	Share of whole-home listings (%)	Capacity-value share (%)	Median price	Multi-listing share (%)
90+ days, 7-night cap	7,335	15.0	28.9	46.6	180	31.5
90+ days, 30-night cap	10,172	20.8	40.0	65.6	180	43.4
90+ days, 90-night cap	10,418	21.3	41.0	67.1	180	43.4
180+ days, 7-night cap	5,102	10.4	20.1	39.1	189	33.5
180+ days, 30-night cap	7,509	15.4	29.6	56.9	185	48.0
180+ days, 90-night cap	7,694	15.7	30.3	58.2	185	47.9
270+ days, 7-night cap	2,967	6.1	11.7	27.3	199	36.4
270+ days, 30-night cap	4,755	9.7	18.7	41.7	195	54.0
270+ days, 90-night cap	4,880	10.0	19.2	42.8	195	54.0

Notes: Each row restricts to entire-home listings and varies the annual availability threshold and minimum-stay cap. The main definition used elsewhere in the paper is 180+ available days and a minimum stay of 30 nights or fewer.

review fields help bound that concern. Among high-availability whole-home candidates, 80.4 percent have at least one review, the median listing has 9.0 reviews and 0.77 reviews per month, and 45.8 percent have at least one review per month. Recency points in the same direction: anchoring to the latest observed review date in the file, 2019-07-08, 65.5 percent of candidate listings have a review within 180 days, and the median reviewed candidate is 22.0 days from its last review. The same row reports a median of 298.0 available days. These numbers do not convert the exposure screen into an occupancy measure, but they show that the candidate margin is not merely a set of dormant listings with open calendars.

Table 3: Listing activity proxies by exposure margin

Listing margin	Listings	Median days available	Reviewed listings (%)	Reviewed last 180 days (%)	Median days since last review	Median reviews/month	Reviews/month ≥ 1 (%)
High-availability whole-home candidates	7,509	298.0	80.4	65.5	22.0	0.77	45.8
Other whole-home listings	17,898	1.0	79.8	43.2	99.0	0.27	28.3
Private rooms	22,319	45.0	79.1	48.8	51.0	0.40	35.2
Shared rooms	1,158	90.0	72.9	49.7	41.5	0.41	36.4

Notes: Review variables are activity proxies from the public listing file, not direct occupancy measures. Review recency is measured relative to the latest observed review date in the file. The candidate margin is entire-home listings with at least 180 available days and a minimum stay of 30 nights or fewer.

The segment is concentrated in the parts of the city with high visitor demand and high residential rents. Table 4 lists the neighborhoods with the largest counts of high-availability whole-home listings. Midtown, Williamsburg, Bedford-Stuyvesant, Harlem, Bushwick, and other amenity-rich or transit-connected areas appear prominently. These are not random neighborhoods. They are places where the visitor willingness-to-pay gradient and the resident willingness-to-pay gradient overlap.

Figure 1 shows the listing composition visually. Figure 2 maps the same logic into neighborhood exposure counts. The figures reinforce the central empirical point: the relevant policy object is not Airbnb activity in general but the high-availability whole-unit segment. A regulation that treats all listings identically will either under-regulate the conversion margin or over-regulate casual home sharing.

Host organization provides additional evidence on the economic margin. Table 5 reports

Table 4: Neighborhoods with largest potential housing-reallocation exposure

Borough	Neighborhood	Listings	Whole-home listings	High-availability whole homes	Median price	Candidate share (%)
Brooklyn	Bedford-Stuyvesant	3710	1591	538	80.000	14.501
Manhattan	Midtown	1545	1100	527	210.000	34.110
Manhattan	Hell's Kitchen	1958	1187	461	168.000	23.544
Manhattan	Upper East Side	1798	1310	392	149.000	21.802
Brooklyn	Williamsburg	3919	1890	388	105.000	9.900
Manhattan	Financial District	744	572	371	200.000	49.866
Manhattan	Upper West Side	1971	1256	331	150.000	16.794
Manhattan	Harlem	2658	1036	272	89.000	10.233
Manhattan	Chelsea	1113	820	242	199.000	21.743
Manhattan	East Village	1853	1170	237	150.000	12.790
Manhattan	Murray Hill	484	398	209	190.000	43.182
Brooklyn	Crown Heights	1564	759	208	85.000	13.299
Brooklyn	Bushwick	2462	678	182	65.000	7.392
Manhattan	West Village	768	653	136	200.000	17.708
Manhattan	Lower East Side	911	506	134	139.000	14.709

Notes: Potential housing-reallocation exposure is defined as an entire-home listing with at least 180 available days and a minimum stay of 30 nights or fewer. The measure is not an estimate of units converted from long-term rental use; it is an exposure screen.

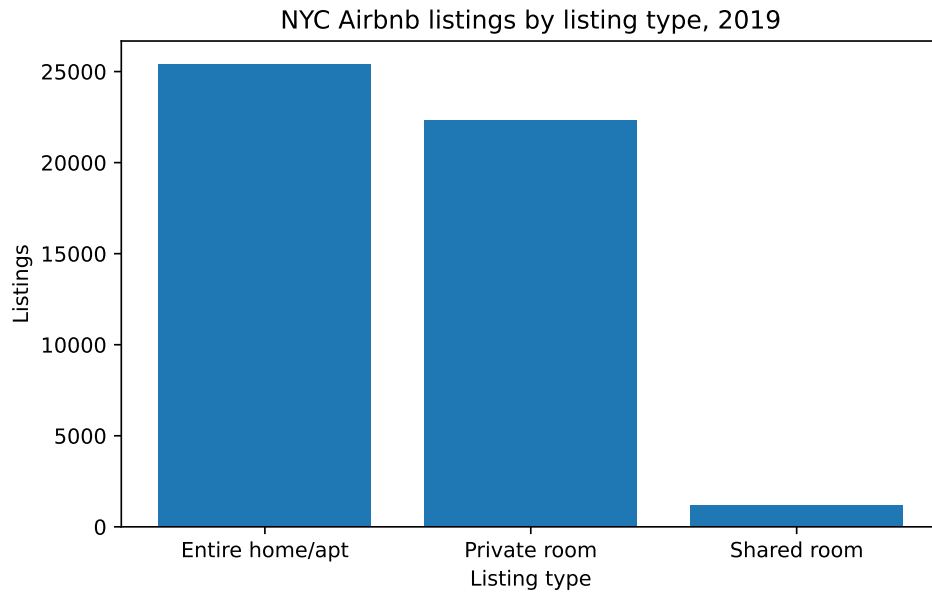


Figure 1: NYC Airbnb listings by listing type, 2019

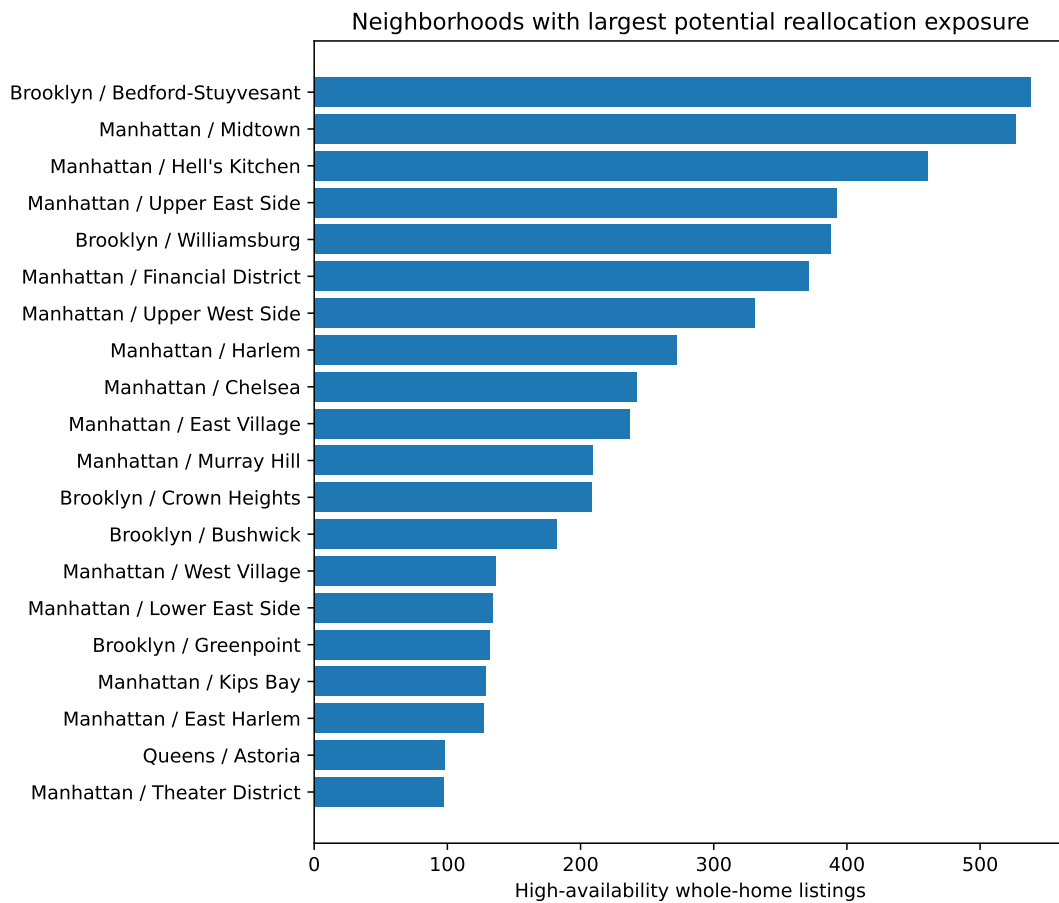


Figure 2: Neighborhoods with largest potential housing-reallocation exposure

that 33.9 percent of listings are associated with hosts who operate multiple listings, and 11.5 percent are associated with hosts who operate at least five listings. The top 100 host identifiers account for 6.2 percent of listings. The market is not monopolized by a handful of hosts, but multi-listing operators are important. This matters because a host with several listings is less likely to be renting out spare capacity in a primary residence. Portfolio supply is the segment most likely to create scarcity rents when housing supply is inelastic.

Table 5: Host concentration in NYC Airbnb listings, 2019

Measure	Value
Unique hosts	37,455
Share of hosts with multiple listings	13.8%
Share of listings by multi-listing hosts	33.9%
Share of listings by top 10 hosts	2.6%
Share of listings by top 100 hosts	6.2%
Listing-count Gini	0.217
Capacity-value share of top 100 hosts	18.6%

Notes: The unit of observation for concentration is the host identifier reported in the listing file.

Table 6 makes the portfolio pattern specific to the reallocation screen. Single-listing hosts account for 52.0 percent of high-availability whole-home candidates, while hosts with at least 10 reported listings account for 22.8 percent. The candidate segment is therefore neither purely professional nor purely casual. It is mixed, but the professional tail is large enough that margin-specific regulation cannot be evaluated as if every listing were occasional home sharing.

Table 6: Host portfolio size and potential housing-reallocation exposure

Host portfolio tier	Hosts	Listings	Share of listings (%)	Whole-home share (%)	High-availability whole homes	Candidate share within tier (%)	Share of candidate segment (%)	Share of candidate capacity value (%)
1 listing	32,301	32,301	66.1	58.8	3,908	12.1	52.0	50.8
2-4 listings	4,640	10,948	22.4	33.9	1,521	13.9	20.3	18.9
5-9 listings	399	2,458	5.0	26.2	371	15.1	4.9	5.3
10+ listings	115	3,177	6.5	64.8	1,709	53.8	22.8	25.0

Notes: Host portfolio tiers use the host listing count reported in the raw Airbnb file. Candidate listings are entire homes with at least 180 available days and a minimum stay of 30 nights or fewer.

Figure 3 plots the Lorenz curve for listing counts across hosts. The curve is bowed but not extreme, indicating that the market combines a large mass of one-listing hosts with a

meaningful professional tail. That mixture is precisely why a single label such as “sharing economy” is insufficient. The same platform hosts genuinely casual transactions and hotel-like supply. A welfare analysis must disaggregate.

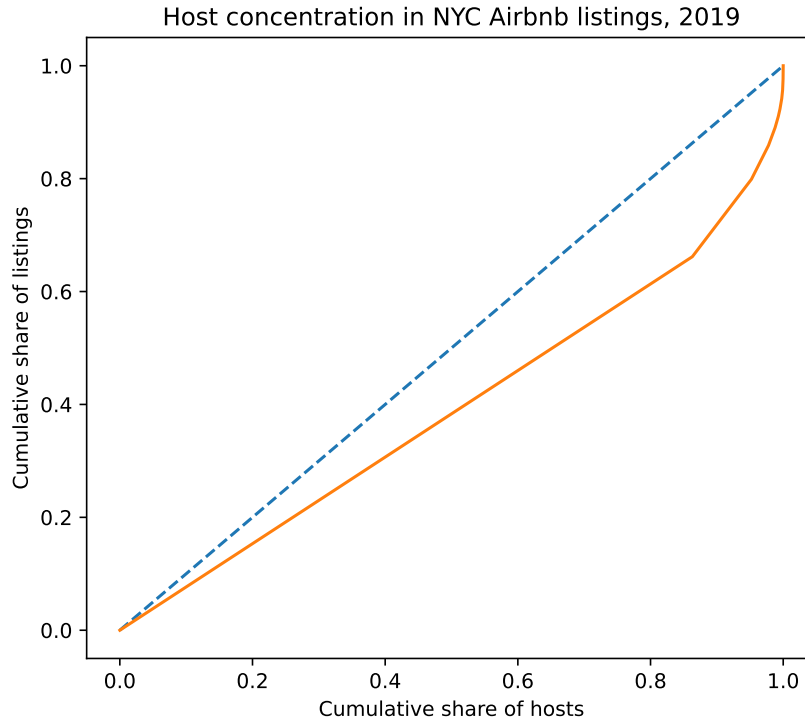


Figure 3: Host concentration in NYC Airbnb listings, 2019

Table 7 reports listing-level price associations. Relative to entire-home listings, private rooms and shared rooms are priced substantially lower, as expected. Multi-listing and professional-host indicators have conditional price associations after controlling for listing type, minimum nights, review history, availability, and borough fixed effects. I do not interpret these coefficients as causal. Instead, they show that host organization and listing form are not merely administrative labels; they are associated with distinct price segments of the platform market.

The listing evidence yields three conclusions. First, the platform market is heterogeneous. A large share of listings plausibly reflects matching surplus, especially private rooms and low-availability listings. Second, the scalable housing-reallocation segment is visible in the data. High-availability whole-home listings are not a theoretical possibility; they are a measurable part of the market. Third, professionalization is not universal but is economically relevant within the exposure segment itself. The welfare decomposition should therefore treat host type and unit type as first-order variables.

Table 7: Listing-level price associations

Variable	Coefficient	SE	Percent association
Private room	-0.803	0.005	-55.193
Shared room	-1.209	0.018	-70.157
Multi-listing host	-0.003	0.006	-0.288
Professional host (5+ listings)	-0.035	0.009	-3.457
Log minimum nights	-0.103	0.003	-9.796
Log reviews	-0.041	0.002	-4.026
Reviews per month	-0.008	0.002	-0.806
Available days	0.001	0.000	0.087

Notes: Outcome is log nightly price, winsorized at the 99th percentile. Specification includes borough fixed effects. Heteroskedasticity-robust standard errors are reported. Coefficients are descriptive associations rather than causal effects.

8 Local Law 18 and the Post-Enforcement Registry

New York City’s Local Law 18 is an example of regulation designed around the distinction between legal home sharing and residential conversion. The law requires short-term rental hosts to register and requires booking platforms to verify a valid registration before processing transactions. It also conditions legal short-term rental activity on rules that restrict entire-apartment rentals for fewer than 30 days and limit short-term rentals to host-resident settings with no more than two guests.

The FY2025 OSE report provides a post-enforcement view of the regulated market. Table 8 reports application statuses during the fiscal year ending June 30, 2025. OSE received 1,582 applications, granted 640, denied 471, and left 107 pending as of the reporting date. The same workbook reports 2,952 active registrations as of June 30, 2025. The number is small relative to the 2019 platform listing market, which contained 48,884 listings. The comparison is not one-to-one because the datasets refer to different dates and legal objects. But that is the point: once the city requires registration and platform verification, the legal short-term rental market becomes a much narrower set of compliant hosts.

Table 9 reports the processing-time diagnostics from the same workbook. Among applications with final determinations, total elapsed time averaged 53.0 days, while time pending with OSE averaged 5.6 days. The gap is economically meaningful because it separates public administrative review from applicant correction and documentation time. Registration therefore screens legal eligibility, but it also imposes a compliance process on otherwise eligible resident hosts.

Table 8: NYC Local Law 18 application statuses, FY2025

Status	Applications
APPLIED FOR	1,582.000
GRANTED	640.000
DENIED	471.000
DENIAL PENDING	304.000
PENDING	107.000
ADDITIONAL INFO REQUESTED	60.000

Notes: The source is the NYC Mayor’s Office of Special Enforcement Short-Term Rental Registration Report covering July 1, 2024 through June 30, 2025.

Table 9: NYC Local Law 18 processing-time diagnostics, FY2025

Measure	Value
Submitted applications	1,582
Applications with final determinations	1,111
Returned for correction	448
Without initial determination	23
Mean total days to final determination	53.0
Mean OSE days to final determination	5.6
Mean total days for correction returns	51.4
Mean OSE days for correction returns	7.2
Mean total days without initial determination	0.5

Notes: Rows are citywide aggregates from the OSE FY2025 processing-time worksheet. Total days include applicant time outside OSE review; OSE days isolate the time pending with the agency.

Table 10 reports citywide denial reasons. OSE reports 4,329 all-time denied applications through June 30, 2025. The largest reported reason is failure to make required corrections, with 2,942 denials, or 68.0 percent of the denied-application total. Rent-regulated units account for 553 denial reasons, or 12.8 percent. Those categories show that registration is not merely a paperwork requirement: it screens applications at the margins relevant to legal occupancy, tenant protection, and administrative verifiability.

Table 10: NYC Local Law 18 citywide denial reasons

Reason	Denied applications	Share of denied applications (%)
Total applications denied	4,329	100.0
Applicant Did Not Make Required Corrections	2,942	68.0
Rent Regulated	553	12.8
Duplicate or Multiple Applications	564	13.0
Building Added to Prohibited Buildings List	189	4.4
Class B Multiple Dwelling - Exempt	37	0.9
Applied-for Address Not Approved for Residential Use	31	0.7
Application Included a Material False Statement	10	0.2
Application Fee Owed	3	0.1

Notes: Rows are all-time citywide denial reasons reported in the FY2025 OSE workbook. Shares divide each reported reason count by the reported total applications denied. Categories are shown as reported by OSE.

Figure 4 shows the distribution of active registrations across council districts. Registration is spatially uneven, reflecting both tourism demand and legal eligibility. The application, processing, and denial tables together clarify the institutional channel: Local Law 18 narrows the legal market not only by requiring hosts to file forms, but by making eligibility verifiable before platform transactions proceed.

The council-district geography makes it possible to compare the pre-enforcement platform market with the post-enforcement legal registry on the same spatial unit. I assign 48,871 2019 listings to current council districts, including 7,507 high-availability whole-home candidates. Table 11 reports the assignment quality for listings, candidate listings, and ACS tract internal points. It separates point-in-polygon matches, unassigned observations, and the ACS nearest-boundary fallback, so the spatial accounting is not an unreported imputed crosswalk. Table 12 reports the districts with the largest pre-enforcement exposure. The citywide ratio is 2.54 assigned high-availability whole-home candidates in 2019 per active registration on June 30, 2025. Figure 5 plots district-level exposure against active registrations. The correlation is only 0.24, which indicates that the legal market after verification is not simply the old high-availability whole-unit segment persisting with registrations.

Raw listing counts are useful but incomplete because council districts differ in housing stock and tenure composition. I therefore normalize exposure by the ACS tract housing denominators assigned to the same district boundaries. Citywide, the assigned high-availability

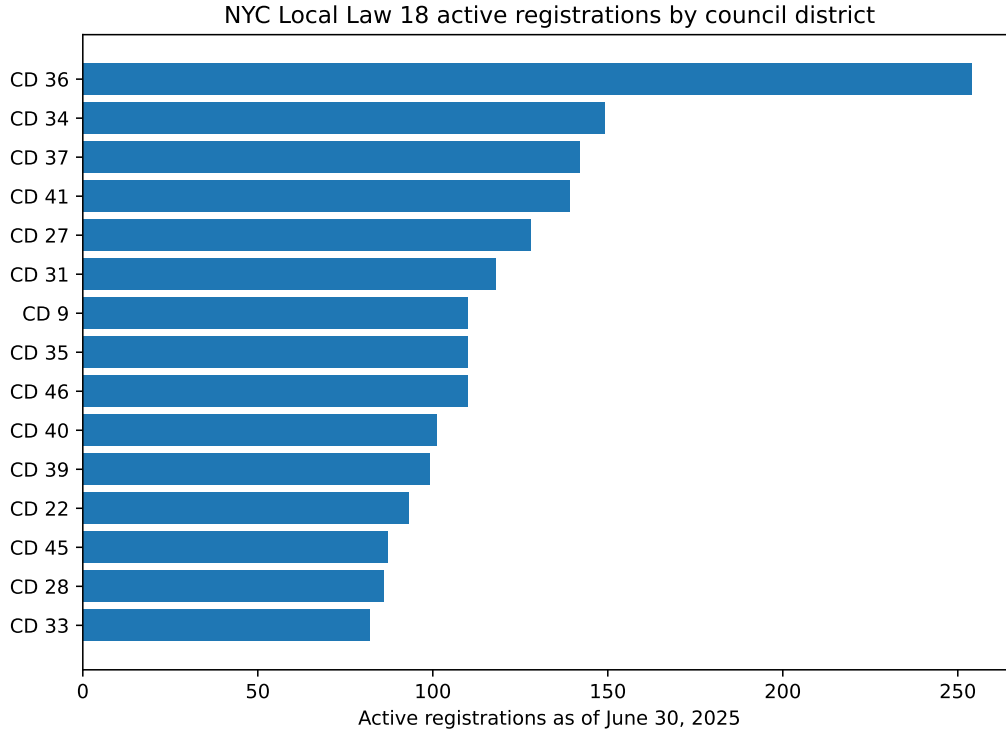


Figure 4: NYC Local Law 18 active registrations by council district

Table 11: Spatial assignment quality for council-district crosswalks

Source object	Assignment method	Observations	Share of source (%)	Assigned observations	Assigned housing units	Max nearest km	Assignment rule
2019 Airbnb listings	Point-in-polygon	48,871	99.973	48,871			Inside council-district polygon
2019 Airbnb listings	Unassigned	13	0.027	0			Outside all council-district polygons
2019 high-availability whole-home candidates	Point-in-polygon	7,507	99.973	7,507			Inside council-district polygon
2019 high-availability whole-home candidates	Unassigned	2	0.027	0			Outside all council-district polygons
ACS 2023 tract internal points	Point-in-polygon	2,317	99.570	2,317	3,629,771		Internal point inside council-district polygon
ACS 2023 tract internal points	Nearest boundary within 1 km	7	0.301	7	20,125	0.076	Nearest council boundary within 1 km
ACS 2023 tract internal points	Unassigned	3	0.129	0	0		No council-boundary match within 1 km

Notes: Listing rows use point-in-polygon assignment from listing coordinates to current council-district polygons. ACS rows use TIGERweb tract internal points; the nearest-boundary fallback assigns only tracts whose internal points fall within one kilometer of a council-district polygon. Unassigned ACS tracts are excluded from district denominators.

Table 12: Council districts with the largest pre-enforcement reallocation exposure

Council district	2019 listings	2019 high-availability whole homes	Candidate share (%)	FY2025 active registrations	2019 candidates per active registration
3	4,033	951	23.6	54	17.61
1	2,994	709	23.7	38	18.66
4	1,996	676	33.9	15	45.07
2	3,279	528	16.1	52	10.15
36	3,313	477	14.4	254	1.88
34	4,438	379	8.5	149	2.54
6	1,641	377	23.0	19	19.84
5	1,620	337	20.8	17	19.82
33	2,534	297	11.7	82	3.62
35	2,353	285	12.1	110	2.59
9	2,167	250	11.5	110	2.27
39	1,624	226	13.9	99	2.28

Notes: Listings are assigned to current New York City Council district boundaries using listing latitude and longitude. The OSE registration counts are active registrations as of June 30, 2025. The comparison is a spatial accounting exercise, not a listing-level panel: the 2019 listing file and FY2025 registry refer to different legal regimes and dates.

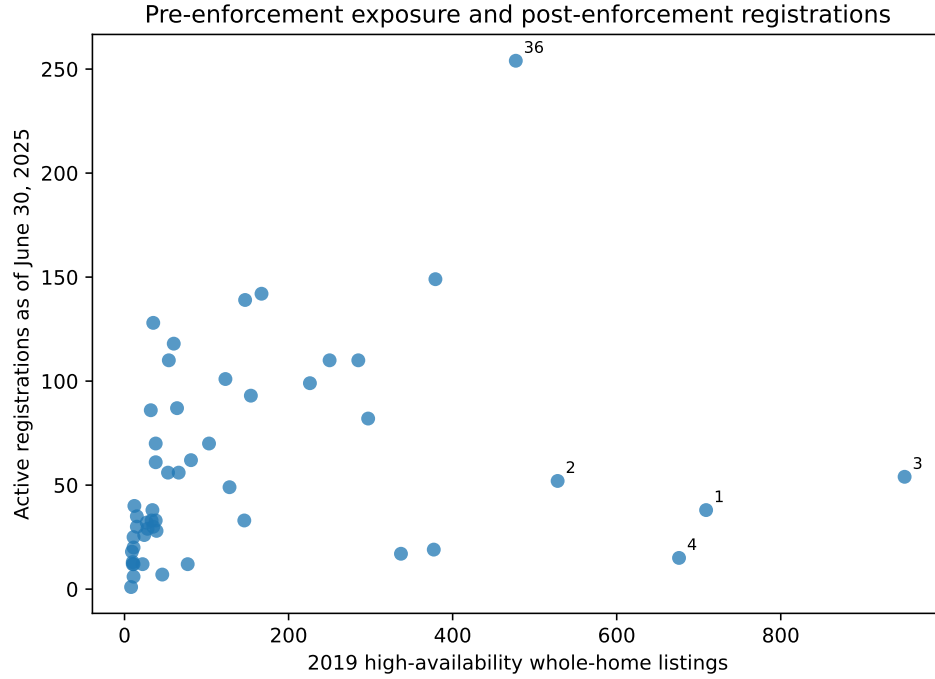


Figure 5: Council-district listing exposure and active registrations

whole-home segment equals 2.06 candidates per 1,000 housing units and 3.37 candidates per 1,000 renter-occupied households. Active registrations equal only 1.33 per 1,000 renter households. Table 13 ranks districts by candidate exposure per renter household. The top districts remain concentrated in Manhattan, where tourist demand overlaps with high rents and dense renter occupancy. Figure 6 plots the normalized pre-enforcement exposure against normalized active registrations. The correlation is 0.05, even lower than the raw-count correlation. This strengthens the interpretation that the post-enforcement legal market is not a scaled-down copy of the pre-enforcement whole-unit segment.

Table 13: Council-district exposure normalized by renter households

Council district	2019 high-availability whole homes	ACS renter households	2019 candidates per 1,000 renter households	FY2025 active registrations	Active registrations per 1,000 renter households	Renter-weighted median gross rent
4	676	48,555	13.92	15	0.31	3,143
3	951	73,386	12.96	54	0.74	2,754
1	709	59,091	12.00	38	0.64	2,291
36	477	54,577	8.74	254	4.65	1,895
2	528	63,368	8.33	52	0.82	2,313
6	377	59,561	6.33	19	0.32	2,757
34	379	61,511	6.16	149	2.42	2,206
5	337	57,814	5.83	17	0.29	2,743
33	297	51,101	5.81	82	1.60	2,775
39	226	42,503	5.32	99	2.33	2,551
35	285	54,554	5.22	110	2.02	2,201
9	250	63,637	3.93	110	1.73	1,476

Notes: ACS denominators are 2019–2023 five-year tract estimates. Tract internal points from TIGERweb are assigned to New York City Council district polygons, then tract housing counts are aggregated to districts. Renter-weighted median gross rent averages tract medians using renter-occupied households as weights.

Vacancy is a sharper denominator for the housing-scarcity margin than total housing

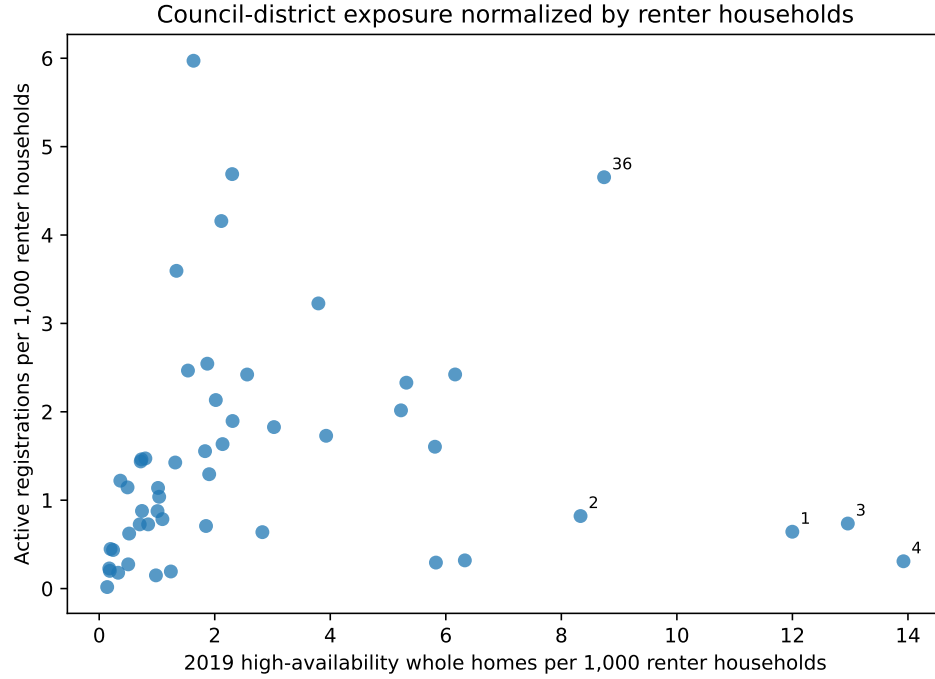


Figure 6: Council-district exposure normalized by renter households

stock, although the public ACS extract used here reports all vacant units rather than vacant units available for rent. Table 14 therefore uses all ACS vacant housing units as a conservative denominator. Citywide, the pre-enforcement high-availability whole-home segment equals 2.23 candidates per 100 vacant housing units, compared with 0.88 active registrations per 100 vacant units after enforcement. Districts with high exposure relative to vacancies are again concentrated in the high-rent, high-tourism core. The vacancy-normalized exposure/registration correlation is 0.56, so the post-enforcement legal registry remains spatially distinct even when the denominator is local vacant housing stock.

Table 14: Council-district exposure relative to ACS vacant housing units

Council district	2019 high-availability whole homes	ACS vacant housing units	2019 candidates per 100 vacant units	ACS vacancy rate (%)	FY2025 active registrations	Active registrations per 100 vacant units	Renter-weighted median gross rent
34	379	3,310	11.45	4.5	149	4.50	2,206
36	477	5,259	9.07	7.1	254	4.83	1,895
1	709	13,406	5.29	14.7	38	0.28	2,291
3	951	18,766	5.07	16.4	54	0.29	2,754
35	285	5,831	4.89	7.4	110	1.89	2,201
33	297	6,825	4.35	8.9	82	1.20	2,775
39	226	5,503	4.11	7.3	99	1.80	2,551
37	167	4,343	3.85	7.0	142	3.27	1,867
2	528	13,916	3.79	14.2	52	0.37	2,313
41	147	4,781	3.07	6.5	139	2.91	1,413
9	250	8,264	3.03	9.9	110	1.33	1,476
4	676	27,359	2.47	25.4	15	0.05	3,143

Notes: The ACS denominator is all vacant housing units, not only vacant units for rent. The ratio is therefore a conservative housing-tightness comparison for short-term-rental exposure: a smaller vacant-for-rent denominator would mechanically produce larger ratios.

A complementary district diagnostic compares posted short-term capacity prices with long-term rent benchmarks. For each high-availability whole-home candidate, I compute

a 30-night capacity price as the winsorized nightly price multiplied by 30, then compare district medians with renter-weighted ACS gross rent. Citywide, the candidate segment has a median nightly price of \$185 and a median 30-night capacity price of \$5,550, compared with a renter-weighted ACS gross-rent benchmark of \$1,887. The 30-night capacity ratio is 2.94, which implies a break-even benchmark of 10.2 booked nights per month. Using annual reported availability, matching annual ACS rent would require 42.7 percent of median available-day capacity. Table 15 reports the districts with the highest ratios among districts with at least 50 candidate listings. These ratios should not be read as realized profit margins: they ignore occupancy, platform fees, taxes, cleaning costs, risk, and seasonality. They do show why the high-availability whole-unit margin can be privately attractive in scarce neighborhoods even when the same physical unit has a long-term residential use.

Table 15: Short-term capacity prices, ACS rents, and break-even occupancy by council district

Council district	2019 high-availability whole homes	Median nightly price	30-night capacity	ACS gross rent	Break-even nights/month	30-night capacity / ACS rent	Annual availability capacity	Annual capacity / annual ACS rent	Break-even annual availability share (%)
8	128	150	4,500	1,291	8.61	3.48	44,845	2.89	34.6
9	250	165	4,950	1,476	8.94	3.35	45,045	2.54	39.3
7	146	195	5,850	1,834	9.40	3.19	54,090	2.46	40.7
1	709	229	6,870	2,291	10.00	3.00	68,456	2.49	40.2
41	147	139	4,170	1,413	10.17	2.95	35,343	2.08	48.0
2	528	225	6,750	2,313	10.28	2.92	64,336	2.32	43.1
31	60	148	4,425	1,617	10.96	2.74	45,298	2.33	42.8
38	81	160	4,800	1,756	10.98	2.73	46,560	2.21	45.3
10	77	150	4,500	1,647	10.98	2.73	45,000	2.28	43.9
3	951	239	7,170	2,754	11.52	2.60	70,168	2.12	47.1
6	377	225	6,750	2,757	12.25	2.45	68,880	2.08	48.0
34	379	180	5,400	2,206	12.26	2.45	50,478	1.91	52.4

Notes: The table compares posted short-term-rental capacity prices with ACS rent benchmarks in districts with at least 50 candidate listings. The 30-night capacity is the winsorized nightly price multiplied by 30. Annual availability capacity is the winsorized nightly price multiplied by reported available days. Break-even values ask how much of posted capacity would match the ACS gross-rent benchmark. These are potential gross capacity measures, not realized revenue, net income, or causal returns.

The district table uses medians, so I also compute the break-even benchmark listing by listing. Each high-availability whole-home listing is assigned the renter-weighted ACS gross rent for its council district. Dividing that benchmark by the listing’s posted nightly price gives the monthly booked nights needed to match long-term rent. Table 16 reports the distribution by host portfolio tier. Across assigned candidate listings, the median break-even benchmark is 12.3 booked nights per month. 16.4 percent of candidates break even within seven booked nights, and 61.6 percent break even within 14 booked nights. The benchmark falls within reported annual availability for 90.6 percent of candidates. This distribution is consistent with the capacity-rent mechanism: the private incentive is not limited to a few extreme listings, although the calculation still abstracts from costs, taxes, fees, seasonality, and actual occupancy.

Table 17 summarizes the same mechanism by ACS gross-rent quartile rather than by top-ranked districts. The highest-rent quartile of council districts contains 65.9 percent of assigned high-availability whole-home candidates, with 7.46 candidates per 1,000 renter households and 3.10 candidates per 100 ACS vacant housing units. Within that quartile, the

Table 16: Listing-level break-even occupancy by host portfolio tier

Host portfolio tier	Candidate listings	Median nightly price	Median ACS gross rent	Median break-even nights/month	p25 break-even nights/month	p75 break-even nights/month	Share at most 7 nights (%)	Share at most 14 nights (%)	Median annual availability share (%)	Share within available days (%)
All candidates	7,507	185	2,291	12.3	8.6	16.6	16.4	61.6	51.3	90.6
1 listing	3,907	180	2,206	12.2	8.2	16.3	18.0	62.5	52.7	89.1
2-4 listings	1,521	170	1,951	12.0	8.1	15.9	17.6	63.8	50.7	91.8
5-9 listings	371	171	2,291	13.4	9.0	19.2	17.5	55.0	51.8	86.3
10+ listings	1,708	208	2,754	12.7	9.5	17.5	11.2	58.8	48.4	93.9

Notes: Each high-availability whole-home listing is assigned the renter-weighted ACS gross-rent benchmark for its council district. Break-even nights divide that rent benchmark by the listing's winsorized posted nightly price. The table describes posted capacity incentives and does not observe realized bookings, host costs, taxes, platform fees, or net income.

median candidate nightly price is \$200, and the listing-level median break-even benchmark is 12.3 booked nights per month. This is not a causal rent-gradient design. It is a mechanism check showing that the reallocation screen is concentrated where both residential rents and visitor prices are high.

Table 17: Council-district exposure by ACS gross-rent quartile

ACS rent quartile	Districts	Renter-weighted ACS gross rent	2019 high-availability whole homes	Share of city candidates (%)	Candidates per 1,000 renter households	Candidates per 100 vacant units	Active registrations per 1,000 renter households	Median candidate nightly price	Median break-even nights/month
Q1 lowest ACS rent	13	1,388	801	10.7	1.22	1.46	0.99	145	9.9
Q2	13	1,689	935	7.1	1.23	1.03	1.55	125	13.7
Q3	12	1,895	1,227	16.3	2.63	1.76	1.82	145	13.1
Q4 highest ACS rent	13	2,508	4,944	65.9	7.46	3.10	1.16	200	12.3

Notes: Rows group the 51 council districts by quartiles of the renter-weighted ACS gross-rent benchmark. District counts, renter households, vacant units, candidate counts, and active registrations are aggregated within quartile. Candidate nightly prices and break-even nights are listing-level medians within the corresponding district quartile. The ACS vacancy denominator is all vacant units.

Table 18 adds a two-dimensional scarcity check. I split districts by whether renter-weighted ACS gross rent is above the citywide renter-weighted benchmark and whether the ACS vacancy rate is below the citywide ACS vacancy rate. High-rent cells jointly contain 75.5 percent of candidate listings. The high-rent, low-vacancy cell alone contains 24.5 percent of candidates, with 4.33 candidates per 1,000 renter households and 3.43 candidates per 100 ACS vacant units. Relative to lower-rent districts that are also below the citywide vacancy rate, the high-rent, low-vacancy cell has 3.66 times as many candidates per renter household and 2.78 times as many candidates per vacant unit. This table is useful because ACS vacancy is not simply low in every high-rent Manhattan district; some high-rent districts have high measured vacancy. The exposure pattern therefore reflects the overlap between visitor demand and high residential rents, with the low-vacancy split showing where that overlap is most directly connected to local housing tightness.

Table 18: Council-district exposure by rent and vacancy scarcity cells

ACS rent/vacancy cell	Districts	Renter-household share (%)	Vacant-unit share (%)	2019 high-availability whole homes	Candidate share (%)	Candidates per 1,000 renter households	Candidates per 100 vacant units	Active registrations per 1,000 renter households	Median break-even nights/month
High rent, low vacancy	11	19.0	15.9	1,836	21.5	4.33	3.43	2.08	13.8
High rent, higher vacancy	8	21.0	42.3	3,835	51.1	8.22	2.69	6.77	11.9
Lower rent, low vacancy	28	52.1	33.1	1,376	18.3	1.18	1.24	1.30	12.4
Lower rent, higher vacancy	4	7.9	8.7	460	6.1	2.63	1.56	1.17	9.5

Notes: High-rent districts have renter-weighted ACS gross rent at or above the citywide renter-weighted benchmark. Low-vacancy districts have ACS vacancy rates below the citywide ACS vacancy rate. The ACS vacancy denominator is all vacant housing units, not only vacant units listed for rent.

Table 19 puts the rent-gradient pattern in regression form. After controlling for ACS

vacancy rates and renter shares, a \$1,000 higher renter-weighted ACS gross-rent benchmark is associated with 4.47 more high-availability whole-home candidates per 1,000 renter households, with a robust standard error of 0.81. The corresponding association is 3.90 candidates per 100 ACS vacant housing units, with a robust standard error of 1.18. By contrast, the same controlled gradient for post-enforcement active registrations is -0.02 per 1,000 renter households. These regressions are cross-sectional descriptive gradients, not estimates of the causal effect of rent levels on listing supply, but they quantify the mechanism implied by the quartile table: the pre-enforcement reallocation screen rises with local rent scarcity while the legal registry does not follow the same gradient.

Table 19: Council-district rent gradients for STR exposure

Outcome	Controls	ACS rent coefficient	Robust SE	p-value	R-squared	Spearman rho	Districts
2019 candidates per 1,000 renter households	No controls	5.60	0.92	0.000	0.55	0.58	51
2019 candidates per 1,000 renter households	Vacancy rate and renter share	4.47	0.81	0.000	0.66	0.58	51
2019 candidates per 100 vacant units	No controls	1.91	0.57	0.001	0.15	0.36	51
2019 candidates per 100 vacant units	Vacancy rate and renter share	3.90	1.18	0.001	0.40	0.36	51
FY2025 active registrations per 1,000 renter households	No controls	-0.22	0.32	0.500	0.01	0.08	51
FY2025 active registrations per 1,000 renter households	Vacancy rate and renter share	-0.02	0.47	0.974	0.14	0.08	51

Notes: Each row is a council-district regression of the listed outcome on renter-weighted ACS gross rent measured in thousands of dollars. Robust HC1 standard errors are reported. Controlled specifications include the ACS vacancy rate and renter share of occupied housing. Spearman rho reports the rank correlation between ACS gross rent and the outcome.

Table 20 asks whether that gradient is an artifact of the main 180-day availability threshold and 30-night minimum-stay cap. I reclassify whole-home exposure using all nine combinations of 90, 180, and 270 available days and 7, 30, and 90 minimum-night caps, then re-estimate the same controlled council-district rent-gradient regression for candidates per 1,000 renter households. The controlled ACS-rent coefficient remains positive across all nine definitions, ranging from 1.16 to 6.22 candidates per 1,000 renter households for a \$1,000 higher ACS gross-rent benchmark. Even under the least concentrated definition in the table, 53.2 percent of candidate listings are in the highest-rent quartile of council districts. This strengthens the mechanism interpretation: the high-rent geography is not produced by one arbitrary exposure cutoff.

Table 21 shows that this spatial pattern should be interpreted as a platform-supply gradient rather than as proof that a single availability cutoff uniquely identifies converted housing. The highest-rent quartile contains 65.9 percent of high-availability whole-home candidates and 68.2 percent of other whole-home listings. Private rooms also skew toward high-rent districts, with 46.5 percent in the highest-rent quartile. The post-enforcement legal registry is different: only 26.1 percent of active registrations are in the highest-rent quartile, and its controlled rent gradient is -0.02 per 1,000 renter households. The comparison sharpens the mechanism. The pre-enforcement platform market was concentrated where visitor demand and residential rents overlap; Local Law 18's legal registry is not simply that

Table 20: Council-district rent-gradient sensitivity across exposure definitions

Availability days	Minimum-night cap	Main	Candidate listings	Share in highest-rent quartile (%)	Controlled ACS-rent gradient	Robust SE	p-value	Districts
90	7		7,334	56.2	3.85	0.90	0.000	51
90	30		10,170	64.3	6.05	1.11	0.000	51
90	90		10,415	64.4	6.22	1.13	0.000	51
180	7		5,101	56.4	2.64	0.64	0.000	51
180	30	Yes	7,507	65.9	4.47	0.81	0.000	51
180	90		7,691	66.0	4.59	0.83	0.000	51
270	7		2,966	53.2	1.16	0.38	0.002	51
270	30		4,753	66.5	2.57	0.54	0.000	51
270	90		4,878	66.8	2.65	0.56	0.000	51

Notes: Each row redefines potential housing-reallocation exposure as an entire-home listing with at least the listed availability threshold and no more than the listed minimum-night cap. The controlled ACS-rent gradient is the coefficient on renter-weighted ACS gross rent, measured in thousands of dollars, in a council-district regression of candidates per 1,000 renter households controlling for ACS vacancy rate and renter share of occupied housing. Robust HC1 standard errors are reported.

same platform geography with fewer listings.

Table 21: Council-district rent-gradient comparison across STR margins

Margin	Citywide count	Per 1,000 renter households	Share in highest-rent quartile (%)	Q4/Q1 rate ratio	Controlled ACS-rent gradient	Robust SE	p-value
High-availability whole homes	7,507	3.37	65.9	6.11	4.47	0.81	0.000
Other whole-home listings	17,892	8.03	68.2	7.66	15.64	2.15	0.000
Private-room listings	22,314	10.02	46.5	3.26	12.44	4.31	0.004
Shared-room listings	1,158	0.52	44.4	2.94	0.37	0.13	0.005
FY2025 active registrations	2,952	1.33	26.1	1.17	-0.02	0.47	0.974

Notes: Rows compare 2019 listing margins with the FY2025 legal registry on the same council-district geography. The Q4/Q1 rate ratio compares each margin per 1,000 renter households in the highest- and lowest-ACS-rent quartiles. The controlled ACS-rent gradient is the coefficient on renter-weighted ACS gross rent, measured in thousands of dollars, in a council-district regression of the margin per 1,000 renter households controlling for ACS vacancy rate and renter share of occupied housing. Robust HC1 standard errors are reported.

Figure 7 plots the same contrast. The left panel shows the upward ACS-rent gradient for pre-enforcement high-availability whole-home exposure normalized by renter households. The right panel shows that active registrations after enforcement do not exhibit the same upward gradient. This visual check is useful because the quartile and regression results are district-level summaries; the figure shows the underlying cross-district pattern rather than only the fitted coefficients.

Because posted nightly prices are not host net revenue, Table 22 repeats the listing-level calculation after stylized net-revenue haircuts. With a 20 percent haircut, the median break-even benchmark rises to 15.3 booked nights per month; 43.1 percent of candidates still break even within 14 booked nights, and 80.2 percent remain within reported annual availability. With a 40 percent haircut, the median rises to 20.4 nights, and the share breaking even within 14 nights falls to 24.0 percent. The sensitivity does not estimate realized profits. It shows how the posted-price incentive changes once gross capacity is scaled toward net revenue.

Local Law 18 also changes the political economy of enforcement. Before platform verification, a city could prohibit certain short-term rentals on paper but face high costs in identifying listings, matching them to addresses, proving violations, and compelling platform

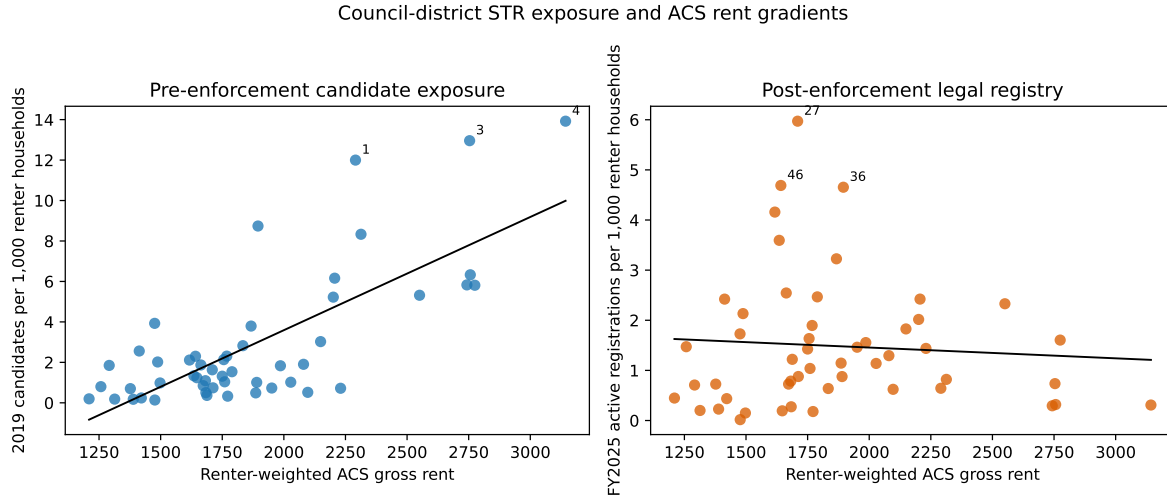


Figure 7: Council-district STR exposure and ACS rent gradients

Table 22: Listing-level break-even sensitivity to net-revenue haircuts

Net-revenue haircut (%)	Candidate listings	Median net nightly price	Median break-even nights/month	p75 break-even nights/month	Share at most 14 nights (%)	Share at most 21 nights (%)	Share at most 30 nights (%)	Median annual availability share (%)	Share within available days (%)
0	7,507	185	12.3	16.6	61.6	88.3	98.3	51.3	96.6
10	7,507	166	13.6	18.5	52.0	83.7	96.7	57.0	86.1
20	7,507	148	15.3	20.8	43.1	75.4	94.7	64.2	80.2
30	7,507	130	17.5	23.8	33.0	65.7	88.5	73.3	72.5
40	7,507	111	20.4	27.7	24.0	52.0	80.4	85.6	61.8

Notes: The haircut rows scale each listing's posted nightly price by one minus the listed haircut before computing the ACS-rent break-even benchmark. The exercise is a stylized sensitivity for platform fees, taxes, cleaning and operating costs, risk, and other leakage from posted gross price to host net revenue. It does not observe realized bookings, realized transaction prices, or the actual cost structure of any listing.

cooperation. Verification requirements move the enforcement margin upstream. The platform must check the legality of a listing before completing a transaction. That design directly targets regulatory rents. If a platform’s scale depends on unregistered or legally ambiguous supply, platform verification reduces the value of that supply by making legality a condition of market access.

The OSE data also reveal administrative costs. Processing time for applications with final determinations averaged 53.0 days in total, but only 5.6 days pending with OSE. The difference reflects applicant correction periods and the complexity of proving legal occupancy. These costs matter for welfare. A strict registration system can reduce illegal conversion but may also burden compliant households. The optimal policy depends on the relative size of the external housing cost and compliance cost. The empirical evidence here suggests that New York chose a high-enforcement equilibrium after observing a large pre-regulation platform market.

9 Housing Market Dynamics Around Restrictions

I next examine housing-market dynamics around restrictive short-term rental policies in Boston, Los Angeles, and New York City. The laws differ in timing, scope, enforcement, and pre-existing market conditions, so the estimand is an aggregate policy-window contrast rather than a law-specific structural parameter. The event study asks whether rent and home-value patterns around restrictions are broadly consistent with the reallocation channel.

Table 23 lists the treated policy windows and raw Zillow changes. The pre/post columns average the 12 months before the coded policy month and the first 12 months beginning with that month. These raw changes are positive for both rents and home values in all three treated metros, which underscores why the analysis cannot rely on simple before-after comparisons. The empirical question is whether treated metros moved differently from comparable untreated metros after absorbing policy-window shocks and pre-policy dynamics.

Table 23: Treated-market policy windows and raw Zillow changes

Market	Policy month	Policy	Mean ZORI, 12 months pre	Mean ZORI, 12 months post	Raw ZORI change (%)	Mean ZHVI, 12 months pre	Mean ZHVI, 12 months post	Raw ZHVI change (%)
Boston, MA	2019-01	Boston STR ordinance	2,326	2,403	3.3	479,726	492,117	2.6
Los Angeles, CA	2019-07	Los Angeles Home-Sharing Ordinance	2,173	2,241	3.1	656,147	670,087	2.1
New York, NY	2023-09	NYC Local Law 18 enforcement	2,974	3,092	4.0	616,214	649,373	5.4

Notes: Pre values average the 12 months before the coded policy month. Post values average the first 12 months beginning with the policy month. These raw pre/post changes are descriptive and do not use the stacked event-study controls or synthetic-control donor weights.

Table 24 reports the policy-event coding used by the Zillow designs. This is part of the identifying object because a metro event study is only as interpretable as its treatment timing. The Boston and Los Angeles policies are coded to the months in which their regulatory

frameworks took effect. New York is coded to September 2023, when platform-verification enforcement began. The omitted baseline is the month before each coded event month, and the second-year post window is the 13–24 month bin used for the main long-horizon estimates. The table also reports the primary treatment margin, which differs across cities: owner-occupancy or primary-residence eligibility in Boston and Los Angeles, and platform verification in New York.

Table 24: Policy-event coding for Zillow designs

Market	Policy month	Omitted baseline month	Primary treatment margin	Coding basis	Second-year post window
Boston, MA	2019-01	2018-12	Owner-occupancy eligibility and registration	Policy framework took effect January 1, 2019; coded to the January 2019 Zillow month.	2020-02–2021-01
Los Angeles, CA	2019-07	2019-06	Primary-residence eligibility and registration	Home-Sharing Ordinance took effect July 1, 2019; coded to the July 2019 Zillow month.	2020-08–2021-07
New York, NY	2023-09	2023-08	Platform verification and registration enforcement	Platform-verification enforcement began September 2023, coded to the September 2023 Zillow month.	2024-10–2025-09

Notes: The event month is the Zillow month to which each local restriction or enforcement milestone is assigned. The omitted baseline month is event month -1 in the stacked event-study design. The full machine-readable codebook, including citation keys and all event windows, is generated at `outputs/validation/policy_event_codebook.csv`.

Table 25 reports a pre-policy donor-pool diagnostic. For each treated market and outcome, it compares the treated metro with untreated large Zillow metros over event months -24 through -2. The table shows that the donor pool is large, but treated metros are not balanced on levels: Boston, Los Angeles, and New York are high-rent or high-value markets relative to the average untreated metro. Pre-policy growth differences are smaller than the level gaps in some windows and larger in others. This diagnostic reinforces the interpretation of the stacked event study as a policy-window design and motivates the city-specific synthetic controls, which match pre-policy trajectories rather than relying on raw level comparability.

Table 25: Pre-policy Zillow balance for treated metros and untreated donor pool

Market	Outcome	Treated pre mean	Control pre mean	Level gap (%)	Treated pre growth (%)	Control pre growth (%)	Growth gap (pp)	Control metros
Boston, MA	Rents (ZORI)	2,285	1,144	99.8	5.7	8.3	-2.5	143
Boston, MA	Home values (ZHVI)	464,071	240,840	92.7	11.9	11.0	0.9	143
Los Angeles, CA	Rents (ZORI)	2,123	1,167	82.0	7.9	7.5	0.4	143
Los Angeles, CA	Home values (ZHVI)	632,137	247,457	155.5	9.7	10.6	-0.8	142
New York, NY	Rents (ZORI)	2,855	1,553	83.8	18.5	12.5	6.0	143
New York, NY	Home values (ZHVI)	605,061	351,757	72.0	9.2	14.0	-4.8	141

Notes: The pre-policy window is event months -24 through -2. Controls are untreated large Zillow metros with complete data in the same pre-policy window. Levels are means over the pre-policy months; growth is the first-to-last-month change within that window. The table is a transparency diagnostic for the donor pool, not a matching or weighting estimator.

Table 26 reports the calendar months behind the post-period bins. This diagnostic is important because the Boston and Los Angeles second-year windows overlap heavily with the COVID-19 housing-market shock: Boston’s 13–24 month bin contains 11 pandemic months, Los Angeles’s contains 12, and New York’s contains none. The pooled second-year rent coefficient therefore cannot be interpreted as an average cleanly separated from pandemic-era dynamics. I keep the pooled estimate because it is informative about policy-window timing across the available restrictive events, but I interpret it jointly with the pretrend,

placebo, leave-one-out, and synthetic-control diagnostics rather than as a self-contained causal estimate.

Table 26: Policy-window calendar diagnostics

Market	Policy month	Post 1–12 months	Pandemic months, post 1–12	Post 13–24 months	Pandemic months, post 13–24
Boston, MA	2019-01	2019-02–2020-01	0/12	2020-02–2021-01	11/12
Los Angeles, CA	2019-07	2019-08–2020-07	5/12	2020-08–2021-07	12/12
New York, NY	2023-09	2023-10–2024-09	0/12	2024-10–2025-09	0/12

Notes: Pandemic months are March 2020 through December 2021. The table documents calendar composition of the event-time bins used in the stacked event-study design. It is a diagnostic for interpreting the pooled post-period estimates, not an additional estimator.

Figures 8 and 9 plot ZORI rent and ZHVI home-value trends in treated metros and the control-metro mean, indexed to January 2018. The figures show the difficulty of causal inference in metropolitan housing markets. Rents and home values move for many reasons: income growth, migration, interest rates, construction, pandemic shocks, and local amenities. New York’s 2023 enforcement occurs after a period of pandemic-era disruption and recovery. Boston and Los Angeles policies occur before the pandemic. This heterogeneity motivates the stacked design and also limits interpretation.

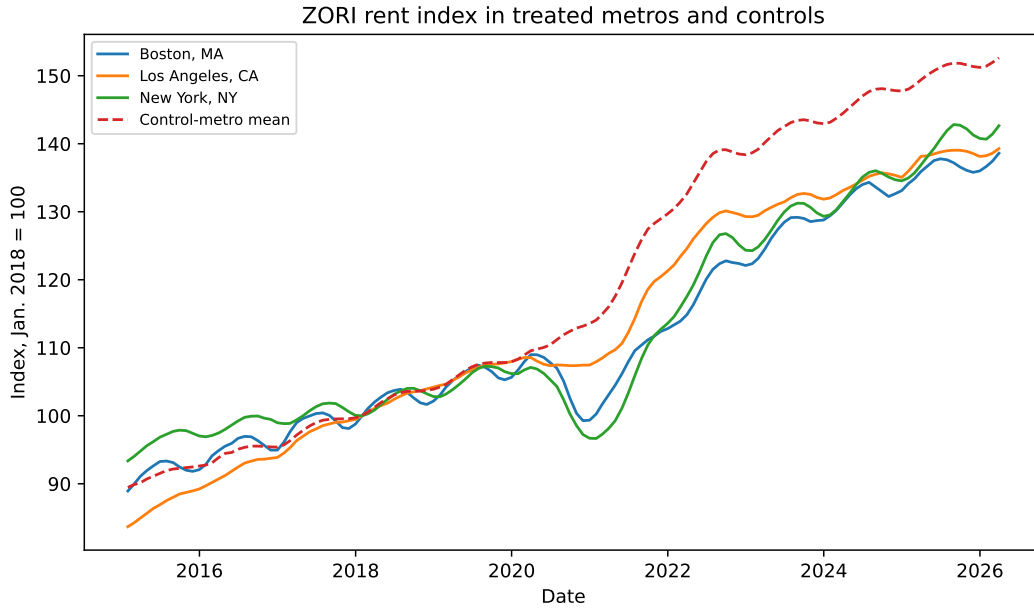


Figure 8: ZORI rent trends in treated metros and controls

Table 27 reports binned event-study estimates. The rent estimates show no large pre-trend in the 24 to 13 months before treatment and a small positive coefficient in months -12 to -2. The month-zero and first-year post coefficients are close to zero. The 13–24 month post coefficient is -3.4 percent, with a standard error of 1.7 percentage points. The home-value

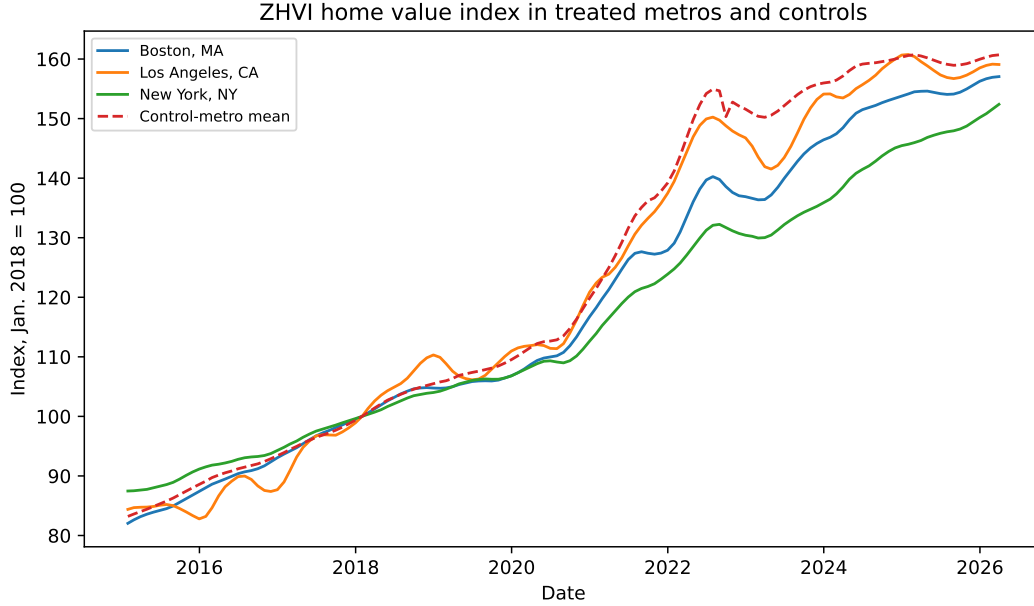


Figure 9: ZHVI home-value trends in treated metros and controls

estimates are less consistent: pre-period coefficients are positive, and the 13–24 month post coefficient is 0.9 percent with a large standard error. These full-sample patterns suggest caution. The rental coefficient is consistent with restrictions relieving some pressure on rents, but its interpretation depends on calendar composition, and the asset-price evidence does not support a clean monotonic story.

Table 27: Stacked event-study estimates around STR restrictions

Outcome	Event-time bin	Effect (%)	SE (pp)	Observations	Clusters
Rents (ZORI)	Pre: -24 to -13	-0.309	1.241	21167	146
Rents (ZORI)	Pre: -12 to -2	0.316	0.642	21167	146
Rents (ZORI)	Event month	0.157	0.165	21167	146
Rents (ZORI)	Post: 1 to 12	-0.115	0.357	21167	146
Rents (ZORI)	Post: 13 to 24	-3.407	1.670	21167	146
Home values (ZHVI)	Pre: -24 to -13	1.135	0.465	21160	146
Home values (ZHVI)	Pre: -12 to -2	1.440	1.064	21160	146
Home values (ZHVI)	Event month	-0.177	0.104	21160	146
Home values (ZHVI)	Post: 1 to 12	0.336	0.886	21160	146
Home values (ZHVI)	Post: 13 to 24	0.908	1.975	21160	146

Notes: Reported effects are percent changes relative to the omitted month -1. The estimation stacks Boston, Los Angeles, and New York City policy/enforcement events and uses untreated large metros as controls. Standard errors are clustered by metro. Estimates are treated as evidence on dynamic patterns, not definitive proof of causal housing-market effects.

Table 28 varies the untreated Zillow control pool by restricting the panel to the largest 100, 150, or 200 metros. The main specification is the top-150 sample. The second-year rent estimate remains negative across these cuts, while the exact magnitude and standard error move with the control pool. The exercise is useful for transparency because the treated cities are large, high-cost metros; it shows that the sign of the main rent pattern is not created solely by one arbitrary size-rank cutoff, but it also reinforces that the pooled coefficient is a design-dependent policy-window estimate rather than a structural constant.

Table 28: Event-study sensitivity to untreated control-pool breadth

Max Zillow size rank	Control metros	Outcome	Event-time bin	Effect (%)	SE (pp)	Observations	Clusters
100	94	Rents (ZORI)	Pre: -12 to -2	0.32	0.64	13,965	97
100	94	Rents (ZORI)	Post: 13 to 24	-3.03	1.67	13,965	97
100	94	Home values (ZHVI)	Pre: -12 to -2	1.42	1.05	13,962	97
100	94	Home values (ZHVI)	Post: 13 to 24	0.85	1.98	13,962	97
150	143	Rents (ZORI)	Pre: -12 to -2	0.32	0.64	21,167	146
150	143	Rents (ZORI)	Post: 13 to 24	-3.41	1.67	21,167	146
150	143	Home values (ZHVI)	Pre: -12 to -2	1.44	1.06	21,160	146
150	143	Home values (ZHVI)	Post: 13 to 24	0.91	1.97	21,160	146
200	186	Rents (ZORI)	Pre: -12 to -2	0.37	0.61	27,467	189
200	186	Rents (ZORI)	Post: 13 to 24	-3.46	1.56	27,467	189
200	186	Home values (ZHVI)	Pre: -12 to -2	1.49	1.06	27,481	189
200	186	Home values (ZHVI)	Post: 13 to 24	0.92	1.94	27,481	189

Notes: Each row re-estimates the same stacked binned event-study specification after restricting the Zillow panel to metros at or above the reported size-rank threshold. The main specification uses the top-150 sample. The table reports the late pre-period bin and second post-year bin because those are most informative about pre-period comparability and the main rent result.

Table 29 tightens the donor comparison in a different way. For each treated policy event, I rank untreated metros by the distance between their normalized pre-policy ZORI and ZHVI trajectories and the treated metro’s normalized trajectory over event months -24 through -2. I then re-estimate the same stacked binned event-study specification using only the closest 20, 50, or 100 untreated metros for each event. With the closest 20 controls per event, the second-year rent estimate is -2.9 percent with a standard error of 0.9 percentage points and 53 metro clusters. The estimate remains negative, but the table should be read as a comparability diagnostic: matching on pre-policy Zillow trajectories reduces reliance on the broad untreated pool without solving endogenous policy adoption or city-specific shocks.

Table 30 splits the same binned design by treated policy event. Boston and Los Angeles account for the negative pooled second-year rent pattern: their event-specific rent gaps are -4.8 and -5.8 percent in the second post-policy year. New York’s corresponding rent gap is positive, at 0.6 percent, after September 2023 enforcement. The home-value rows are also mixed, including a large positive New York second-year value gap. The table therefore strengthens the heterogeneity claim while limiting the causal interpretation. It shows where the pooled rent result comes from, but it does not turn each single-city stack into a separately

Table 29: Event-study sensitivity to pre-policy matched control metros

Matched controls/event	Distinct controls	Outcome	Event-time bin	Effect (%)	SE (pp)	Observations	Clusters
20	50	Rents (ZORI)	Pre: -12 to -2	0.27	0.30	3,087	53
20	50	Rents (ZORI)	Post: 13 to 24	-2.91	0.86	3,087	53
20	50	Home values (ZHVI)	Pre: -12 to -2	1.15	0.66	3,087	53
20	50	Home values (ZHVI)	Post: 13 to 24	0.79	1.55	3,087	53
50	102	Rents (ZORI)	Pre: -12 to -2	0.30	0.38	7,497	105
50	102	Rents (ZORI)	Post: 13 to 24	-3.44	1.14	7,497	105
50	102	Home values (ZHVI)	Pre: -12 to -2	1.38	0.78	7,497	105
50	102	Home values (ZHVI)	Post: 13 to 24	0.58	1.60	7,497	105
100	127	Rents (ZORI)	Pre: -12 to -2	0.30	0.53	14,847	130
100	127	Rents (ZORI)	Post: 13 to 24	-3.41	1.38	14,847	130
100	127	Home values (ZHVI)	Pre: -12 to -2	1.42	0.94	14,847	130
100	127	Home values (ZHVI)	Post: 13 to 24	0.84	1.76	14,847	130

Notes: For each treated policy event, untreated metros are ranked by the distance between their normalized pre-policy ZORI and ZHVI trajectories and the treated metro's normalized trajectory over event months -24 through -2. Each row re-estimates the same stacked binned event-study specification after keeping only the closest untreated metros for each event. This is a comparability diagnostic, not a matching estimator that solves endogenous policy adoption.

powered quasi-experiment.

Table 30: Event-specific binned event-study estimates

Market	Outcome	Event-time bin	Effect (%)	Control metros	Clusters
Boston, MA	Rents (ZORI)	Pre: -12 to -2	1.65	143	144
Boston, MA	Rents (ZORI)	Post: 13 to 24	-4.84	143	144
Boston, MA	Home values (ZHVI)	Pre: -12 to -2	0.52	143	144
Boston, MA	Home values (ZHVI)	Post: 13 to 24	-2.82	143	144
Los Angeles, CA	Rents (ZORI)	Pre: -12 to -2	0.40	143	144
Los Angeles, CA	Rents (ZORI)	Post: 13 to 24	-5.82	143	144
Los Angeles, CA	Home values (ZHVI)	Pre: -12 to -2	4.08	143	144
Los Angeles, CA	Home values (ZHVI)	Post: 13 to 24	0.17	143	144
New York, NY	Rents (ZORI)	Pre: -12 to -2	-1.08	143	144
New York, NY	Rents (ZORI)	Post: 13 to 24	0.57	143	144
New York, NY	Home values (ZHVI)	Pre: -12 to -2	-0.24	143	144
New York, NY	Home values (ZHVI)	Post: 13 to 24	5.55	143	144

Notes: Each panel stacks only the named treated metro against untreated large metros, excluding the other treated metros from its control pool. Reported bins are the late pre-period and the second post-policy year. Because each event-specific stack has one treated metro, the table reports effects as heterogeneity diagnostics and leaves inference to the pooled, placebo, and synthetic-control diagnostics.

Table 31 decomposes the pooled binned estimator itself. I residualize the outcome and binned treatment indicators with respect to the stack-by-metro and stack-by-month fixed effects, compute the observation weights for each pooled binned coefficient, and sum weighted residualized outcomes by treated-policy stack. The second-year rent coefficient receives negative contributions from Boston and Los Angeles and a small positive contribution from New York. The residualized weight shares are approximately balanced across the three event

stacks, so the pooled negative rent result is not an artifact of silently overweighting one treated city. It is the weighted average of heterogeneous treated-city paths, with opposing signs in the New York enforcement window.

Table 31: Treated-market contributions to pooled binned event-study estimates

Market	Outcome	Event-time bin	Pooled effect (%)	Event-specific effect (%)	Contribution to pooled log gap (pp)	Share of pooled log gap (%)	Residualized weight share (%)	Treated months
Boston, MA	Rents (ZORI)	Pre: -12 to -2	0.32	1.65	0.54	172.5	33.3	11
Los Angeles, CA	Rents (ZORI)	Pre: -12 to -2	0.32	0.40	0.13	42.6	33.3	11
New York, NY	Rents (ZORI)	Pre: -12 to -2	0.32	-1.08	-0.36	-115.0	33.3	11
Boston, MA	Rents (ZORI)	Post: 13 to 24	-3.41	-4.84	-1.65	47.7	33.3	12
Los Angeles, CA	Rents (ZORI)	Post: 13 to 24	-3.41	-5.82	-2.00	57.7	33.3	12
New York, NY	Rents (ZORI)	Post: 13 to 24	-3.41	0.57	0.19	-5.4	33.3	12
Boston, MA	Home values (ZHVI)	Pre: -12 to -2	1.44	0.52	0.17	12.2	33.3	11
Los Angeles, CA	Home values (ZHVI)	Pre: -12 to -2	1.44	4.08	1.33	93.3	33.3	11
New York, NY	Home values (ZHVI)	Pre: -12 to -2	1.44	-0.24	-0.08	-5.5	33.3	11
Boston, MA	Home values (ZHVI)	Post: 13 to 24	0.91	-2.82	-0.95	-105.4	33.3	12
Los Angeles, CA	Home values (ZHVI)	Post: 13 to 24	0.91	0.17	0.06	6.1	33.3	12
New York, NY	Home values (ZHVI)	Post: 13 to 24	0.91	5.55	1.80	199.3	33.3	12

Notes: The table decomposes the residualized stacked fixed-effect estimator. For each outcome and event-time bin, observation weights from the corresponding binned coefficient are multiplied by residualized outcomes and summed by treated-policy stack. Contributions sum to the pooled log coefficient up to numerical tolerance. Shares can be negative or exceed 100 percent when treated-market contributions have opposite signs.

Table 32 repeats the binned event-study specification after dropping March 2020 through December 2021 from every stack. This exercise directly addresses the calendar overlap in Table 26. The second-year rent coefficient changes from -3.4 percent in the full sample to 1.6 percent, with a standard error of 0.4 percentage points, and is identified by only 13 treated event-months after the exclusion. The sensitivity does not create a better causal design; it shows that the pooled second-year estimate is partly a statement about policy windows that pass through the pandemic period. I therefore place more weight on calendar sensitivity, city-level heterogeneity, and concordance across diagnostics than on the exact pooled magnitude.

Table 32: Event-study sensitivity to excluding pandemic calendar months

Outcome	Event-time bin	Full effect (%)	Full SE (pp)	Full treated months	No-pandemic effect (%)	No-pandemic SE (pp)	No-pandemic treated months
Rents (ZORI)	Post: 1 to 12	-0.11	0.36	36	0.19	0.23	31
Rents (ZORI)	Post: 13 to 24	-3.41	1.67	36	1.63	0.35	13
Home values (ZHVI)	Post: 1 to 12	0.34	0.89	36	0.57	1.05	31
Home values (ZHVI)	Post: 13 to 24	0.91	1.97	36	4.95	1.39	13

Notes: The no-pandemic specification drops March 2020 through December 2021 from every stack and then re-estimates the same binned event-study specification with stack-by-metro and stack-by-month fixed effects. Treated months count treated event-by-calendar-month observations available in each bin after the corresponding sample restriction.

Table 33 makes the pre-period evidence explicit. The monthly-lead joint tests reject zero pretrends for both outcomes, with p-values that round to 0.000 for rents and 0.000 for home values in the -12 to -2 window. The economic magnitude differs across outcomes. Rent leads average only a few tenths of a percentage point in the late pre-period, whereas home-value leads are larger and more visibly trending. This diagnostic narrows the interpretation of the event study: the rent estimates are useful as policy-window evidence when read alongside the

listing, registry, placebo, and synthetic-control checks, but the design should not be read as a standalone parallel-trends proof.

Table 33: Stacked event-study pretrend diagnostics

Outcome	Pre-period window	Leads	Mean lead (%)	Largest absolute lead (%)	Largest lead month	Cluster-robust F	p-value	Observations	Clusters
Rents (ZORI)	Monthly leads: -24 to -2	23	-0.01	1.24	-24	27.66	0.000	21167	146
Rents (ZORI)	Monthly leads: -12 to -2	11	0.32	0.66	-11	15.04	0.000	21167	146
Home values (ZHVI)	Monthly leads: -24 to -2	23	1.28	2.26	-7	151.51	0.000	21160	146
Home values (ZHVI)	Monthly leads: -12 to -2	11	1.44	2.26	-7	149.83	0.000	21160	146

Notes: The table reports joint tests that the monthly pre-treatment event-study coefficients equal zero. Month -1 is the omitted category. The F statistics use the same stack-by-metro and stack-by-month fixed effects and metro-clustered covariance matrix as the monthly event-study estimates.

Table 34 reports the linear-pretrend-adjusted diagnostic. For rents, the fitted pre-period slope extrapolates to a small positive second-year gap of 0.2 percent. Subtracting that extrapolated drift leaves a second-year rent gap of -3.6 percent, with an adjusted standard error of 0.9 percentage points. The rent pattern therefore is not mechanically explained by the simple linear pretrend visible in the monthly leads. For home values, the adjustment produces a 2.5 percent second-year gap with a 2.4 percentage-point standard error. That result reinforces the paper’s treatment of the value evidence as inconclusive: the pre-period movement is too large for a clean asset-price interpretation.

Table 34: Linear-pretrend-adjusted event-study gaps

Outcome	Post-period window	Pretrend slope (pp/month)	Unadjusted mean gap (%)	Extrapolated pretrend (%)	Trend-adjusted gap (%)	Adjusted SE (pp)	Observations	Clusters
Rents (ZORI)	Post: 1 to 12	0.01	-0.11	0.08	-0.19	0.87	21167	146
Rents (ZORI)	Post: 13 to 24	0.01	-3.41	0.21	-3.61	0.92	21167	146
Home values (ZHVI)	Post: 1 to 12	-0.08	0.34	-0.60	0.94	1.09	21160	146
Home values (ZHVI)	Post: 13 to 24	-0.08	0.91	-1.55	2.49	2.35	21160	146

Notes: The diagnostic fits a linear differential trend to the monthly pre-treatment event-study coefficients from months -24 to -2, anchored at the omitted month -1, and subtracts the extrapolated trend from post-period monthly gaps. It is a sensitivity check for pre-period drift, not a substitute for exogenous policy assignment.

Table 35 re-estimates the binned event study after omitting one treated metro at a time. This check asks whether the full-sample second-year rent pattern is mechanically driven by a single policy event. The estimates remain in the same direction across omissions, although magnitudes change because each two-event stack is necessarily thin. That pattern is useful for describing the full-sample event-study estimate, while the calendar-sensitivity table and synthetic controls preserve the broader causal caveats about metro-level policy timing.

Table 36 provides a complementary placebo exercise. I assign the three policy dates to metros that are untreated in the stacked design and compute the same second-year post-period contrast relative to the month before the placebo event. This check does not create random policy assignment, but it asks whether the observed treated-metro effect is common among large untreated metros when the same calendar dates are imposed mechanically. The observed rent effect falls in the lower tail of the placebo distribution, while remaining inside

Table 35: Leave-one-policy-event-out event-study estimates

Omitted event	Outcome	Pre: -12 to -2 (%)	Pre SE (pp)	Post: 13 to 24 (%)	Post SE (pp)	Clusters
Boston, MA	Rents (ZORI)	-0.342	0.530	-2.681	2.323	145
Boston, MA	Home values (ZHVI)	1.900	1.496	2.823	1.860	145
Los Angeles, CA	Rents (ZORI)	0.272	0.960	-2.175	1.956	145
Los Angeles, CA	Home values (ZHVI)	0.144	0.280	1.281	2.918	145
New York, NY	Rents (ZORI)	1.023	0.437	-5.335	0.443	145
New York, NY	Home values (ZHVI)	2.288	1.229	-1.336	1.098	145

Notes: Each row re-estimates the stacked event study after excluding one treated metro. Reported coefficients are percent effects for the -12 to -2 pre-period bin and the 13 to 24 month post-period bin. Standard errors are clustered by metro.

the range of placebo estimates; this reinforces a cautious interpretation rather than a sharp randomization-inference claim.

Table 36: Placebo policy-date distribution among untreated metros

Outcome	Observed effect (%)	Placebo mean (%)	Placebo SD (pp)	Placebo p5 (%)	Placebo p95 (%)	Left-tail share	Centered two-sided share	Placebos
Rents (ZORI)	-3.407	0.049	3.095	-4.939	4.904	0.114	0.221	429
Home values (ZHVI)	0.908	0.073	3.702	-6.325	5.968	0.594	0.779	429

Notes: The placebo exercise assigns the Boston, Los Angeles, and New York policy dates to each metro untreated in the stacked design. For each placebo metro-date pair, the table computes the 13 to 24 month post-period change relative to the month before the placebo event using the mean of the remaining untreated metros as a donor benchmark. The left-tail share is the fraction of placebo effects no larger than the observed effect.

Table 37 aligns the placebo exercise more closely with the main estimator. For each of 500 fixed-seed draws, I assign the three real policy dates to three distinct untreated metros, exclude Boston, Los Angeles, and New York from the placebo control pools, and re-estimate the binned stacked event study with the same fixed effects. The second-year rent estimate has a left-tail share of 0.044 and a centered two-sided share of 0.056 across 500 placebo triplets. The home-value estimate is not similarly unusual. This strengthens the timing evidence for the rent result while preserving the central caveat: the exercise evaluates whether the observed stacked pattern is uncommon under reassigned policy dates, not whether the actual cities adopted restrictions as-if randomly.

Table 37: Stacked event-study placebo-date diagnostics

Outcome	Event-time bin	Observed effect (%)	Placebo mean (%)	Placebo p5 (%)	Placebo p95 (%)	Left-tail share	Centered two-sided share	Placebos	Minimum clusters
Rents (ZORI)	Pre: -12 to -2	0.32	-0.01	-0.98	1.02	0.710	0.596	500	143
Rents (ZORI)	Post: 13 to 24	-3.41	-0.01	-3.25	2.86	0.044	0.056	500	143
Home values (ZHVI)	Pre: -12 to -2	1.44	0.02	-1.46	1.47	0.946	0.110	500	143
Home values (ZHVI)	Post: 13 to 24	0.91	-0.02	-3.43	3.42	0.668	0.642	500	143

Notes: The diagnostic uses a fixed-seed set of placebo triplets: for each draw, the three real policy dates are assigned to three distinct untreated metros, Boston, Los Angeles, and New York are excluded from the placebo control pools, and the same binned stacked event-study specification is re-estimated with stack-by-metro and stack-by-month fixed effects. The left-tail share is the fraction of placebo estimates no larger than the observed stacked estimate for the same outcome and event-time bin.

Table 38 reports the city-specific synthetic-control check. The average second-year rent gap across treated metros is -2.7 percent, with an average pre-period RMSPE of 0.34 percentage points. The average masks heterogeneity. Boston and Los Angeles show negative second-year rent gaps relative to their synthetic controls, while New York shows a positive rent gap after September 2023 enforcement. The pooled event-study pattern is therefore driven more by the earlier Boston and Los Angeles policy windows than by a clean New York rent decline after Local Law 18. That heterogeneity is consistent with the paper’s cautious interpretation: short-term-rental restrictions can matter, but the aggregate metro response depends on timing, enforcement, and contemporaneous housing-market shocks.

Table 38: City-specific synthetic-control robustness estimates

Treated metro	Outcome	Pre RMSPE (pp)	Post 1-12 gap (%)	Post 13-24 gap (%)	Selected donors	Largest donor weights
Boston, MA	Rents (ZORI)	0.49	0.56	-3.47	20	Chicago, IL (0.41); New Haven, CT (0.33); Albany, NY (0.26)
Boston, MA	Home values (ZHVI)	0.07	-1.30	-2.71	20	Worcester, MA (0.60); Providence, RI (0.22); Lancaster, PA (0.08); Madison, WI (0.05); Minneapolis, MN (0.03)
Los Angeles, CA	Rents (ZORI)	0.14	-0.83	-6.74	20	Lancaster, PA (0.31); Fort Wayne, IN (0.22); San Diego, CA (0.15); Asheville, NC (0.13); Charleston, SC (0.12)
Los Angeles, CA	Home values (ZHVI)	1.22	2.22	3.72	20	San Francisco, CA (0.88); Miami, FL (0.07); North Port, FL (0.05)
New York, NY	Rents (ZORI)	0.39	0.63	2.03	20	San Diego, CA (0.50); Portland, ME (0.24); Savannah, GA (0.24); Worcester, MA (0.03)
New York, NY	Home values (ZHVI)	0.09	2.50	5.26	20	Anchorage, AK (0.28); Peoria, IL (0.21); Corpus Christi, TX (0.15); Washington, DC (0.12); Gulfport, MS (0.08)

Notes: Each treated metro is normalized to the month before policy/enforcement. Donors are untreated large metros with complete data in the event window. The procedure screens to the 20 closest pre-period donor trajectories and then estimates nonnegative weights that sum to one using months -24 to -2. Gaps are treated-minus-synthetic differences.

Table 39 opens the donor-weight construction behind the summary estimates. The synthetic controls are not unreported black-box averages: each treated metro-outcome pair is built from the screened donor pool with nonnegative weights that sum to one, and the largest weights are visible in the table. The donor composition also clarifies why I treat the exercise as a robustness diagnostic rather than as a definitive counterfactual. Some synthetic controls rely on a small number of high-weight donors, which improves pre-period fit but leaves the post-period comparison exposed to donor-specific shocks.

Table 40 adds placebo and rank diagnostics to the synthetic-control exercise. For each policy date and outcome, I assign the same date to untreated metros and re-estimate the screened synthetic-control procedure. The Los Angeles rent gap is unusual relative to placebo gaps: the directional tail share is 0.007 and the RMSPE-ratio rank share is 0.049. Boston’s rent gap is closer to the edge of the placebo distribution but less extreme, while New York’s positive rent gap is not rare among placebo metros. These diagnostics make the robustness evidence sharper. They support the claim that at least some restrictions are associated with lower rent paths, especially Los Angeles, but they also rule out a uniform treated-city story.

Figures 10 and 11 plot the monthly synthetic-control gaps. The rent figure shows downward second-year gaps for Boston and Los Angeles and an upward gap for New York. The home-value figure is mixed, with an average second-year gap of 2.1 percent and a range from -2.7 to 5.3 percent. This reinforces the broad conclusion from the stacked diagnostics: the rent

Table 39: Largest synthetic-control donor weights

Treated metro	Outcome	Rank	Donor metro	Weight (%)
Boston, MA	Rents (ZORI)	1	Chicago, IL	40.9
Boston, MA	Rents (ZORI)	2	New Haven, CT	33.0
Boston, MA	Rents (ZORI)	3	Albany, NY	26.1
Boston, MA	Home values (ZHVI)	1	Worcester, MA	59.7
Boston, MA	Home values (ZHVI)	2	Providence, RI	21.8
Boston, MA	Home values (ZHVI)	3	Lancaster, PA	8.3
Los Angeles, CA	Rents (ZORI)	1	Lancaster, PA	30.6
Los Angeles, CA	Rents (ZORI)	2	Fort Wayne, IN	22.5
Los Angeles, CA	Rents (ZORI)	3	San Diego, CA	15.0
Los Angeles, CA	Home values (ZHVI)	1	San Francisco, CA	88.1
Los Angeles, CA	Home values (ZHVI)	2	Miami, FL	6.9
Los Angeles, CA	Home values (ZHVI)	3	North Port, FL	5.0
New York, NY	Rents (ZORI)	1	San Diego, CA	49.5
New York, NY	Rents (ZORI)	2	Portland, ME	24.3
New York, NY	Rents (ZORI)	3	Savannah, GA	23.5
New York, NY	Home values (ZHVI)	1	Anchorage, AK	27.8
New York, NY	Home values (ZHVI)	2	Peoria, IL	21.1
New York, NY	Home values (ZHVI)	3	Corpus Christi, TX	15.1

Notes: The table reports up to the three largest nonzero donor weights for each treated metro and outcome. The full selected-donor weight vector, including zero and near-zero selected donors, is generated at `outputs/tables/table17c_synthetic_control_weights.csv`.

Table 40: Synthetic-control placebo and rank diagnostics

Treated metro	Outcome	Observed gap (%)	Placebo p5 (%)	Placebo p95 (%)	Directional tail share	Two-sided centered share	RMSPE-ratio rank share	Placebos
Boston, MA	Rents (ZORI)	-3.47	-3.78	3.87	0.077	0.147	0.308	143
Boston, MA	Home values (ZHVI)	-2.71	-6.00	4.30	0.106	0.234	0.106	141
Los Angeles, CA	Rents (ZORI)	-6.74	-4.00	4.35	0.007	0.021	0.049	142
Los Angeles, CA	Home values (ZHVI)	3.72	-4.07	4.26	0.078	0.177	0.894	141
New York, NY	Rents (ZORI)	2.03	-3.60	3.99	0.203	0.427	0.427	143
New York, NY	Home values (ZHVI)	5.26	-4.19	3.43	0.021	0.035	0.014	141

Notes: Each row compares the treated metro synthetic-control gap with placebo gaps generated by assigning the same policy date to untreated metros and re-estimating the screened synthetic-control procedure. The directional tail share is the left-tail share for negative observed gaps and the right-tail share for positive observed gaps. The RMSPE-ratio rank share is the fraction of placebo units with a post/pre RMSPE ratio at least as large as the treated metro.

evidence is more aligned with the reallocation mechanism than the asset-price evidence, but neither design supports a precise structural estimate.

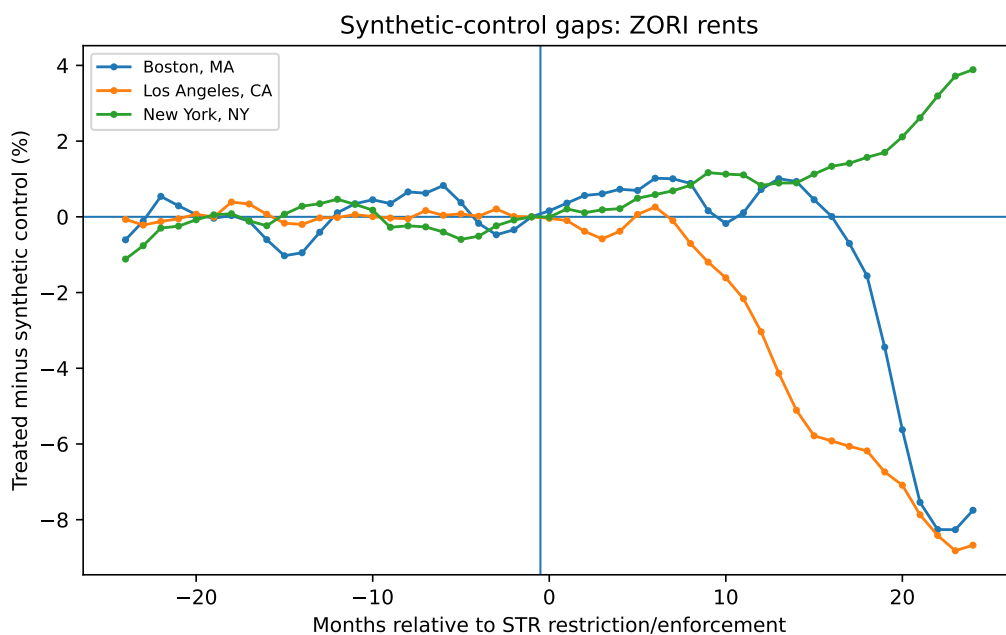


Figure 10: Synthetic-control gaps: ZORI rents

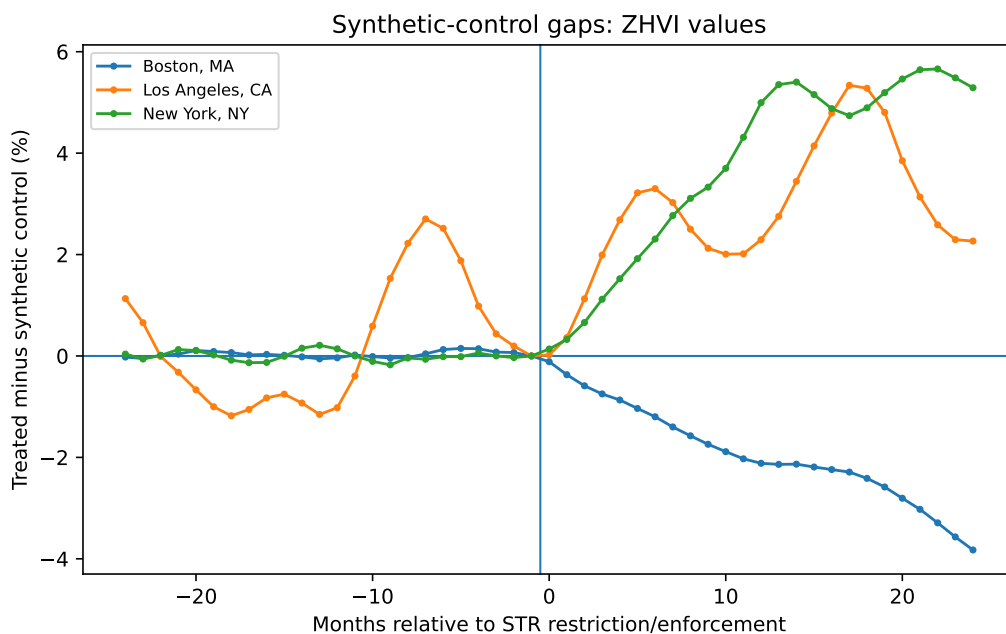


Figure 11: Synthetic-control gaps: ZHVI home values

Figures 12 and 13 report monthly dynamic estimates. The full-sample rent event study

shows estimates drifting downward in the second post year. The home-value event study is noisier and exhibits pre-period movement. This difference is economically plausible. A restriction that returns marginal units to the long-term rental market may affect rents before it affects sale prices, especially if owners and investors expect regulation to be revised or if the relevant units are a small share of the ownership market. It is also possible that the estimates are contaminated by other shocks, especially pandemic-era shocks in the earlier policy windows. The most defensible interpretation is that the rent evidence is suggestive and heterogeneous and the value evidence is inconclusive.

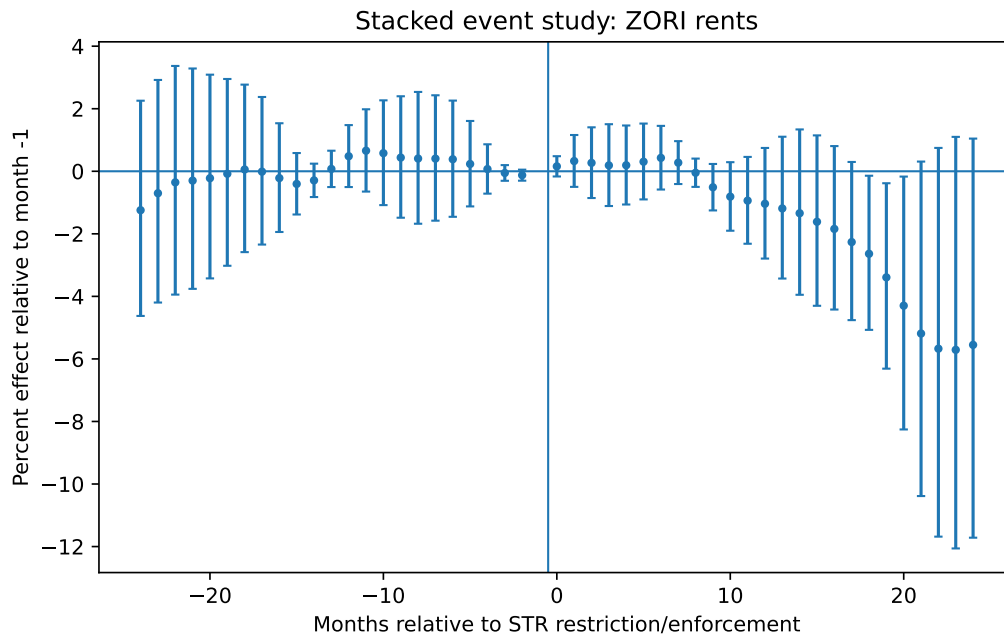


Figure 12: Stacked event study: ZORI rents

The event-study and synthetic-control results align with the paper’s bounded thesis. They do not establish that short-term rental restrictions solve housing affordability. The estimated rent changes are modest relative to the scale of housing-cost growth in constrained markets, and the city-specific synthetic controls show meaningful heterogeneity. They do show that restrictions are not irrelevant in aggregate data. In the context of the listing evidence, the results support a model in which the marginal effect of regulation comes through the high-availability whole-unit segment rather than through casual home sharing.

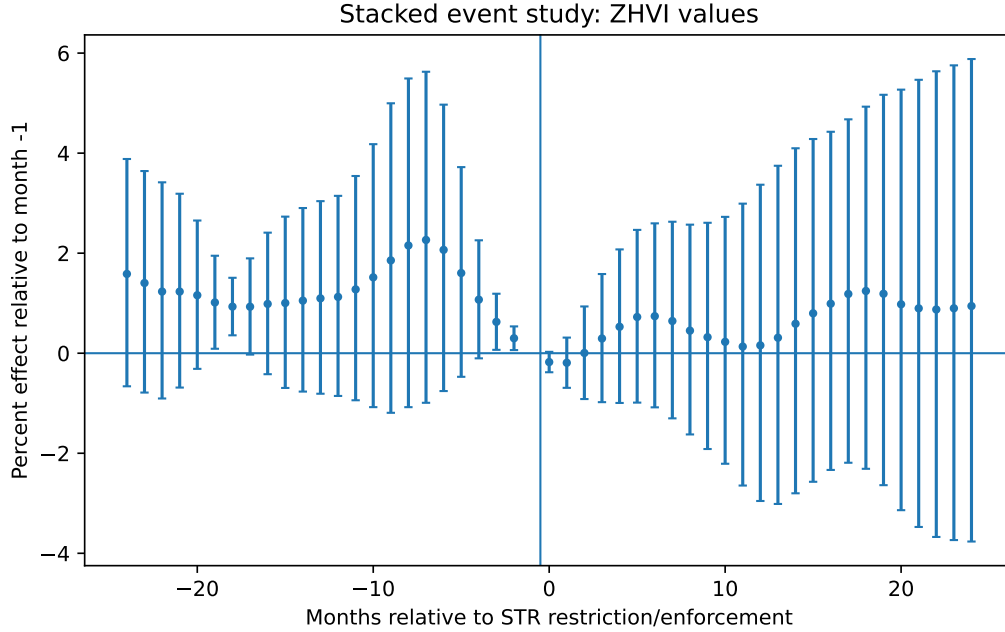


Figure 13: Stacked event study: ZHVI home values

10 Welfare Interpretation

The welfare implication of the evidence depends on the margin. Consider a private room in an owner-occupied apartment. The host's opportunity cost may be low, the resident housing stock is unchanged, and the traveler obtains a cheaper or more differentiated lodging option. The main externalities are neighborhood nuisance, tax compliance, and safety. In this case, platform surplus is close to matching surplus. Regulation should focus on registration, tax remittance, and nuisance rules rather than prohibition.

Now consider an entire apartment available most of the year in a high-rent neighborhood. The opportunity cost is not idle capacity. It is the long-term rental value of the unit plus any option value to the owner. If the short-term rental return exceeds the long-term return because visitors have high willingness to pay and platform matching costs are low, the host's private return includes a housing-scarcity rent. In a market with inelastic housing supply, the conversion imposes costs on renters through a smaller long-term rental stock. The platform may still create traveler surplus and host surplus, but the distributional incidence changes. Renters who do not transact on the platform bear part of the cost.

The New York listing data show both margins. Private rooms and low-availability listings are numerous. So are high-availability entire-home listings. The OSE registration data show how one city attempted to separate them legally. The metro event-study and synthetic-control results suggest that restrictions are associated with lower rent growth in some policy windows,

although the evidence is not decisive enough to assign a precise causal dollar value. The welfare accounting is therefore qualitative but disciplined by observed magnitudes.

A useful policy metric is the ratio of external housing cost to platform capacity value. Let Δr_m be the estimated rent change attributable to short-term rental activity in market m , N_m^R the number of renter households exposed to the change, and CV_m the gross capacity-value of short-term rental supply. Then

$$\Omega_m = \frac{\Delta r_m \times N_m^R}{CV_m} \quad (9)$$

measures the renter-cost burden per dollar of platform capacity value. This paper does not estimate Ω_m because the public data used here do not provide tract-level renter counts linked to listing-level treatment variation and causal rent effects. But the formula clarifies what a complete welfare estimate should report. A platform market can generate large private value with small external cost in an elastic housing market, and small private value with large external cost in a tightly constrained neighborhood.

The decomposition also reframes the debate over platform wealth. The question is not whether a founder, investor, or host can legally make money from a platform. The question is what economic mechanism generated the return. If the return comes from reducing search costs and enabling transactions that use idle capacity, it is innovation surplus. If the return comes from converting scarce residential units into transient accommodations while externalizing rent pressure, it is at least partly scarcity rent. If the return depends on delayed enforcement or legal ambiguity, it includes regulatory rent. These categories are not moral labels; they are mechanisms that imply different policy responses.

11 Threats to Identification and Robustness

Several threats are central. First, listing exposure is not observed conversion. An entire-home listing with high availability may be a second home, a unit temporarily vacant for sale, or a legal owner-occupied arrangement. Conversely, a listing with lower availability may still affect the long-term rental market if it prevents a room or unit from being leased. The exposure screen should therefore be interpreted as a necessary empirical partition, not as a definitive classification of illegal or harmful units.

Second, platform data are selected. The AB_NYC_2019 file records listings visible in a particular period and does not observe unlisted units, direct bookings, occupancy, or realized transaction prices. Availability is reported by the platform and may reflect host calendar choices rather than true capacity. The capacity-value measure is therefore an upper-bound proxy for potential gross value, not realized revenue.

Third, policy adoption is endogenous. Cities regulate short-term rentals because they face housing pressure, tourism pressure, neighborhood complaints, or political mobilization. Treated cities are not randomly selected. The event-study design controls for common shocks and fixed differences across stacked units, but it cannot rule out time-varying local shocks. This is particularly important for New York because Local Law 18 enforcement occurs after the pandemic reshaped both rental and ownership markets.

Fourth, treatment is heterogeneous. Boston, Los Angeles, and New York did not adopt identical rules. Boston restricts eligible units and requires registration. Los Angeles restricts home sharing to primary residences, limits days absent extended approval, and excludes rent-stabilized units. New York’s Local Law 18 imposes registration and platform verification under a legal environment that already restricted many entire-apartment rentals. A single stacked event-study coefficient averages across these institutional differences.

Fifth, the metro scale is coarse. Housing-market effects of short-term rental regulation are likely neighborhood-specific. A citywide or metro-level average can dilute effects concentrated in tourist neighborhoods. The council-district crosswalk improves the spatial accounting for New York but still does not create a causal microgeographic design because the listing file is from 2019 and the registration report is from FY2025. The ideal design would compare census tracts or blocks near jurisdictional borders, exploit building-level eligibility, or use platform-verification enforcement timing at the listing level. The present evidence is therefore best read as aggregate and district-level accounting evidence on broad dynamics.

The replication package includes several checks. The event-study tables report pre-period bins, policy-window calendar diagnostics, broad and matched control-pool sensitivity, pandemic-excluded calendar sensitivity, formal monthly-lead diagnostics, and linear-pretrend-adjusted post-period gaps. The leave-one-event-out table tests whether a single policy window drives the pooled estimate, the placebo-date tables compare observed effects with mechanically assigned dates and pooled placebo triplets in untreated metros, and the synthetic-control tables report pre-fit diagnostics, city-specific gaps, placebo distributions, and RMSPE-ratio ranks. The price regression uses robust standard errors and borough fixed effects. The OSE parser extracts official workbook totals and writes them separately from listing and Zillow data. The analysis pipeline fails if raw data files are missing and produces a validation report documenting that no artificial observations are generated. These checks do not eliminate the identification concerns, but they make the empirical claims transparent and replicable.

12 External Validity and Policy Counterfactuals

The results travel across markets only through the margins identified by the framework. In a low-rent market with elastic construction, abundant vacancies, and limited tourism demand, the welfare cost of short-term rental conversion may be small. A platform may mostly monetize spare capacity, second homes, or temporary vacancies. In that environment, strict restrictions could destroy matching surplus with limited renter benefit. In a high-rent, low-vacancy market with strong visitor demand and many whole-unit listings, the same platform technology can have a different incidence. The supply response is weak, the visitor demand pool is large, and the long-term rental market bears the opportunity cost. The proper policy implication is therefore not national uniformity but local calibration.

Three counterfactual policies are useful for interpreting the evidence. The first is *laissez-faire*. Under *laissez-faire*, any residential unit can be listed as a short-term rental if the private return exceeds the host's private cost. This policy maximizes the platform's ability to discover high-value transactions. It also creates the strongest risk that private returns include housing-scarcity rents and regulatory arbitrage rents. *Laissez-faire* is most defensible when housing supply is elastic, listings are predominantly spare rooms or low-availability homes, and nuisance externalities are modest.

The second counterfactual is a broad ban on short-term rentals. A ban protects residential supply but also eliminates transactions that use genuine excess capacity. The welfare loss from a ban is larger when many listings are private rooms, owner-occupied homes, or occasional rentals. The New York listing data show why this matters. The market contains a large private-room segment as well as a large entire-home segment. Treating both as identical would ignore economically meaningful heterogeneity. A broad ban can be rational when enforcement of more targeted rules is impossible or when illegal conversion is pervasive, but it is not the first-best response in the decomposition.

The third counterfactual is margin-specific regulation. This is the approach suggested by the evidence. Registration reduces anonymity. Platform verification increases enforcement probability. Primary-residence rules preserve home sharing while reducing commercial conversion. Night caps distinguish occasional use from year-round visitor accommodation. Prohibitions on rent-regulated, subsidized, or affordable units protect housing categories where the social objective is explicitly residential occupancy. Taxes can recover part of the public cost where transactions remain legal. These instruments correspond directly to the surplus components in the model.

The registry evidence from New York illustrates both the promise and the cost of margin-specific regulation. A much smaller legal market after enforcement suggests that verification

can be binding. The fiscal-year application counts also show administrative costs: applications are granted, denied, held pending additional information, or left pending. The mean total processing time reported in the workbook is 53.0 days among final decisions, while the mean OSE processing component is 5.6 days. A verification regime is therefore not costless. It shifts part of the platform's scale decision into a public administrative process. The welfare question is whether the reduction in housing-scarcity and regulatory rents justifies the compliance and administrative costs.

The decomposition provides a way to evaluate that question without treating all hosts as identical. If most denied applications involve rent-regulated or otherwise ineligible units, enforcement is plausibly protecting housing stock that public policy already intended to reserve for residents. If denied applications are mostly technical errors by otherwise eligible resident hosts, enforcement may be destroying matching surplus. The OSE report's denial categories are therefore not just administrative details; they are evidence about which surplus component regulation is removing. The workbook records 553 all-time denial reasons involving rent-regulated units, which points to the relevance of housing-protection rules in the regulatory process.

A final external-validity issue is the platform's own response. Regulation does not only remove listings. It may change host behavior, platform screening, prices, minimum stays, and the composition of legal supply. Some short-term rental demand may shift to hotels, to legal hosted stays, to mid-term rentals, or to informal channels. Some hosts may return units to the long-term rental market, while others may sell, leave units vacant, or seek other uses. The event study captures reduced-form housing-market dynamics, not the full general equilibrium response. For welfare, the direction of reallocation matters. A unit returned to long-term rental supply has a different effect from a unit shifted into owner occupancy or vacancy.

These counterfactuals also clarify the relation between short-term rental policy and housing-supply reform. STR restrictions cannot substitute for building more housing in constrained cities. If zoning and permitting rules keep supply inelastic, removing a small reallocation pressure will not solve affordability. But supply reform and STR regulation address different margins. Supply reform shifts the long-run housing supply curve. STR regulation changes the short-run allocation of existing units between residents and visitors. The welfare case for STR regulation is strongest during the period in which housing supply is inelastic and reform is incomplete.

13 Discussion

The results imply that short-term rental policy should be designed around economic margins rather than treating the platform market as a single activity. A blanket ban would eliminate matching surplus from spare rooms and low-availability listings. A laissez-faire policy would allow the high-availability whole-unit segment to monetize scarce housing without internalizing renter costs. The economically coherent middle ground is margin-specific regulation: registration, tax compliance, platform verification, primary-residence rules, night caps, and stronger restrictions for rent-regulated, subsidized, or non-owner-occupied units.

The results also speak to platform governance. Platform firms often argue that they merely facilitate voluntary exchange. That claim is incomplete when the exchange reallocates scarce local assets and when nonparticipants bear costs. The platform's matching technology changes the opportunity cost of residential space. By expanding the visitor demand pool, it can raise the private return to short-term use above the long-term residential return. The external effect is not created by the platform alone; it is created by the interaction of platform technology with housing scarcity. But the interaction is precisely what policy must evaluate.

A common counterargument is that short-term rentals are too small relative to the total housing stock to matter. That argument can be correct at the metro level and wrong at the neighborhood margin. Housing markets clear locally. A few thousand units can matter in a tight submarket even if they are small relative to a metropolitan stock. The right empirical comparison is not total listings divided by total housing units; it is high-availability whole-unit listings relative to vacant rental supply in the neighborhoods where they concentrate, and the private short-term capacity price relative to the long-term rent benchmark. The ACS data permit a conservative council-district vacancy comparison using all vacant housing units as the denominator and a renter-weighted gross-rent benchmark for capacity-price comparisons. They do not isolate vacant-for-rent units, identify which vacancies are actually available to long-term renters, or observe realized short-term rental revenue, so these tables should be read as housing-tightness and incentive diagnostics rather than definitive measures of displaced rental supply or profit.

Another counterargument is that housing unaffordability is primarily caused by zoning and insufficient construction, not short-term rentals. This paper agrees. The central housing-scarcity problem in high-cost cities is supply inelasticity. Short-term rentals are a secondary amplifier, not the root cause. But an amplifier can still matter. In an elastic market, conversion into short-term rentals induces new construction or reallocation with small rent effects. In an inelastic market, the same conversion has larger distributional consequences. Thus, short-term rental policy should complement, not substitute for, broader housing supply

reform.

14 Conclusion

This paper develops and applies a surplus-decomposition approach to short-term rental platforms. The central distinction is between matching surplus and scarcity rent. Platforms can create real value by matching travelers with underused housing. They can also create private returns by redirecting scarce residential units toward transient demand. The empirical task is to distinguish those margins.

The empirical results document that New York City’s 2019 Airbnb market contained both casual and professionalized segments. Entire homes or apartments were 52.0 percent of listings, 7,509 listings met a high-availability whole-home screen, and 48.0 percent of that candidate segment was associated with multi-listing hosts. The highest-rent quartile of council districts contains 65.9 percent of assigned candidates, and a \$1,000 higher ACS gross-rent benchmark is associated with 4.47 more candidates per 1,000 renter households after controlling for vacancy rates and renter shares. The median posted 30-night capacity price for the candidate segment is 2.94 times the renter-weighted ACS gross-rent benchmark, and the listing-level median needs 12.3 booked nights per month to match its district benchmark. A 20 percent net-revenue haircut raises that listing-level median to 15.3 nights. This helps explain the private incentive to allocate scarce units toward transient demand. New York’s FY2025 registry under Local Law 18 records only 2,952 active registrations as of June 30, 2025, showing how platform verification and registration can sharply narrow the legal market. The council-district crosswalk shows that the post-enforcement legal registry is spatially distinct from the pre-enforcement high-availability whole-home segment, including after normalizing by ACS renter households and vacant housing units. A stacked event study around restrictions in Boston, Los Angeles, and New York finds treated-metro rents -3.4 percent lower 13–24 months after treatment in the full sample, but the coefficient changes to 1.6 percent after excluding pandemic calendar months and is then based on only 13 treated event-months. The housing-market evidence is therefore a bounded policy-window diagnostic, not a precise pooled causal estimate.

The evidence supports a bounded conclusion. Short-term rental platforms are neither pure innovation nor pure extraction. They are mixed institutions whose welfare effects depend on the unit type, host type, market tightness, and regulatory environment. The platform surplus earned from spare-room matching is economically different from the surplus earned by converting high-demand residential units into visitor accommodation. Policy should preserve the former while pricing, limiting, or prohibiting the latter when housing supply is inelastic

and renter externalities are large.

References

- Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of california’s tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010.
- Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2):495–510, 2015.
- Kyle Barron, Edward Kung, and Davide Proserpio. The effect of home-sharing on house prices and rents: Evidence from airbnb. *Marketing Science*, 40(1):23–47, 2021.
- Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economics and Statistics*, 106(5):1179–1198, 2024.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.
- City of Boston. Short-term rental ordinance and registration requirements. Municipal ordinance and city guidance on short-term rentals, 2018.
- Clément de Chaisemartin and Xavier D’Haultfoeuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020.
- Liran Einav, Chiara Farronato, and Jonathan Levin. Peer-to-peer markets. *Annual Review of Economics*, 8:615–635, 2016.
- Chiara Farronato and Andrey Fradkin. The welfare effects of peer entry: The case of airbnb. *American Economic Review*, 112(6):1782–1817, 2022.
- Miquel-Àngel Garcia-López, Jordi Jofre-Monseny, Rodrigo Martínez-Mazza, and Mariona Segú. Do short-term rental platforms affect housing markets? evidence from airbnb in barcelona. *Journal of Urban Economics*, 119:103278, 2020.
- Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021.
- Joseph Gyourko, Albert Saiz, and Anita Summers. A new measure of the local regulatory environment for housing markets: The wharton residential land use regulatory index. *Urban Studies*, 45(3):693–729, 2008.

- Chang-Tai Hsieh and Enrico Moretti. Housing constraints and spatial misallocation. *American Economic Journal: Macroeconomics*, 11(2):1–39, 2019.
- Hans R. A. Koster, Jos van Ommeren, and Nicolas Volkhausen. Short-term rentals and the housing market: Quasi-experimental evidence from airbnb in los angeles. *Journal of Urban Economics*, 124:103356, 2021.
- Los Angeles Housing Department. Home-sharing ordinance. City guidance on Los Angeles Home-Sharing Ordinance No. 185931, 2019.
- New York City Airbnb Open Data. Ab_nyc_2019 public listing file. Public CSV file mirrored in the alexeygrigorev/datasets GitHub repository, 2019.
- New York City Mayor’s Office of Special Enforcement. Short-term rental registration law and platform verification enforcement. NYC Local Law 18 registration and enforcement guidance, 2023.
- New York City Mayor’s Office of Special Enforcement. Short-term rental registration report, july 1, 2024 to june 30, 2025. Public workbook, Local Law 18 reporting, 2025.
- NYC Open Data and New York City Department of City Planning. City council districts. Public GeoJSON boundary file, accessed June 2026, 2026.
- Jean-Charles Rochet and Jean Tirole. Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4):990–1029, 2003.
- Marc Rysman. The economics of two-sided markets. *Journal of Economic Perspectives*, 23(3):125–143, 2009.
- Albert Saiz. The geographic determinants of housing supply. *Quarterly Journal of Economics*, 125(3):1253–1296, 2010.
- Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.
- U.S. Census Bureau. American community survey 2019–2023 five-year detailed tables summary file. Public table-based Summary File, accessed June 2026, 2024a.
- U.S. Census Bureau. Tigerweb acs 2023 census tracts. Public TIGERweb tract internal-point service, accessed June 2026, 2024b.
- Zillow Research. Housing data: Zillow observed rent index and zillow home value index. Public CSV time-series data, accessed May 2026, 2026.

A Replication Package

The replication package accompanying this manuscript contains all data files used in the empirical analysis, code to parse and clean the data, scripts to reproduce the tables and figures, and a validation report. The core command is `make all`, which runs the analysis pipeline, compiles the manuscript, writes output fingerprints, and executes the validation tests. The pipeline reads the raw listing file, Zillow metro files, OSE workbook, council-district GeoJSON, ACS tract housing extract, and TIGERweb tract internal-point extract; writes processed datasets; estimates the listing, spatial-accounting, event-study, placebo, and synthetic-control models; generates all figures and tables; writes manuscript macros; and records validation diagnostics. The validation directory, `outputs/validation`, includes:

- a raw-data inventory with SHA-256 hashes, byte counts, source URLs, and row, feature, or sheet diagnostics for every raw input;
- a source-code inventory with hashes for analysis code, scripts, tests, configuration files, manuscript source files, and replication documentation;
- a software-environment inventory with the Python runtime, platform, key package versions, and external build commands visible to the workflow;
- a policy-event codebook that records the treated-market event months, source citation keys, coding basis, treatment margins, and event-window definitions used by the Zillow designs;
- `result_macro_source_map.csv`, which links every generated manuscript statistic to its validation metric and enforces that generated result macros are cited in `paper/main.tex`;
- `manuscript_artifact_manifest.csv`, which hashes every LaTeX table input, result macro file, and figure included by `paper/main.tex`;
- `manuscript_figure_source_map.csv`, which links each included figure to the generated data file or files used to draw it; and
- `manuscript_table_source_map.csv`, which links each included LaTeX table to the exact generated CSV source in `outputs/tables`.

The master workflow additionally writes a submission-artifact checklist that fingerprints the compiled manuscript, bibliography, submission statements, replication documentation, data manifest, and validation files, as well as a generated-output inventory with SHA-256 hashes and byte counts for processed datasets, generated tables, figures, validation files, LaTeX table inputs, result macros, and the compiled manuscript PDF. No artificial observations are generated by the pipeline.

B Additional Institutional Details

Boston’s policy event is coded as January 2019, when the city’s short-term rental ordinance took effect. Los Angeles is coded as July 2019, when the Home-Sharing Ordinance took effect. New York is coded as September 2023, when Local Law 18 platform-verification enforcement began. The event-study windows are limited to 24 months before and after each policy month. Other treated metros are excluded from the control group within each stack. The generated file `outputs/validation/policy_event_codebook.csv` records these event-month choices, source citation keys, treatment-margin notes, omitted baseline months, and pre/post event-window ranges.